



Workshop Manual

Crafter 2017 ➤

TGE 2017 ➤

Auxiliary heater

Edition 01.2019



List of Workshop Manual Repair Groups

Repair Group

00 - Technical data

82 - Supplementary heating

Technical information should always be available to the foremen and mechanics, because their careful and constant adherence to the instructions is essential to ensure vehicle road-worthiness and safety. In addition, the normal basic safety precautions for working on motor vehicles must, as a matter of course, be observed.



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00 – Technical data

1 Safety information

(VRL012476; Edition 01.2019)

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⇒ ["1.2 Safety regulations for working on fuel supply system", page 2](#)

⇒ ["1.3 Safety measures when working on vehicles with a start/stop system", page 3](#)

⇒ ["1.4 Safety precautions when using testers and measuring instruments during a road test", page 3](#)

⇒ ["1.5 Safety precautions when working on the cooling system", page 3](#)

⇒ ["1.6 Safety precautions when working on vehicles with auxiliary/supplementary heater", page 3](#)

⇒ ["1.7 Safety precautions when working on SCR system", page 4](#)

1.1 Safety precautions during work on fuel system

Danger of fire caused by escaping fuel

When the battery is connected and the driver door opens, the door contact switch activates the fuel pump. Escaping fuel can ignite and cause a fire.

- Disconnect voltage supply to fuel pump before opening the fuel system.

Risk of injury from highly pressurised fuel.

The fuel system is pressurised. Injury from fuel spray possible.

Before opening the fuel system:

- Wear protective goggles.
- Wear protective gloves.
- To release pressure, wrap a clean cloth around the connection and carefully loosen the connection.



Note

- ◆ *When working on a fuel system whose fuel lines have been disconnected, observe the applicable rules for cleanliness as well as safety precautions ⇒ Rep. gr. 00 ; Safety instructions .*
- ◆ *Air in the fuel supply to the auxiliary heater can lead to malfunction when the auxiliary heater is in combustion mode.*
- ◆ *After repair work has been performed on the fuel tank or the fuel delivery unit, the fuel take-off pipe must be filled with fuel. Otherwise, malfunction may occur from air drawn in by the metering pump - V54- when the auxiliary heater is switched on for the first time.*

1.2 Safety regulations for working on fuel supply system

Risk of injury from highly pressurised fuel.

The fuel system is pressurised. Injury from fuel spray possible.

Before opening the fuel system:

- Wear protective goggles.
- Wear protective gloves.
- To release pressure, wrap a clean cloth around the connection and carefully loosen the connection.

Danger of fire caused by escaping fuel

When the battery is connected and the driver door opens, the door contact switch activates the fuel pump. Escaping fuel can ignite and cause a fire.

- Disconnect voltage supply to fuel pump before opening the fuel system.

Observe fill level of fuel tank

- ◆ Fuel tank must not be filled completely. To find out how much fuel may remain in the fuel tank, refer to the corresponding work procedure.
- ◆ Drain fuel tank if necessary ⇒ Fuel supply system - diesel engines; Rep. gr. 20 ; Fuel tank; Draining fuel tank .
- ◆ Before starting work, the extraction hose of a switched-on exhaust gas extraction system must be laid near to the assembly opening of the fuel tank.
- ◆ The exhaust extraction system is used to extract fuel vapours.
- ◆ If no exhaust gas extraction system is available, a radial fan with a displacement greater than 15 m³/h can be used providing that motor is not in air flow.

To prevent personal injuries and destruction of the injection and glow plug system, observe the following

- ◆ Always switch off the ignition before connecting or disconnecting electrical wiring of injection and glow plug system or connecting/disconnecting tester cables.



- ◆ Wash the engine only with the ignition switched off.
- ◆ Read event memory of engine control unit using ⇒ Vehicle diagnostic tester if electrical connectors have been disconnected and engine was started.

1.3 Safety measures when working on vehicles with a start/stop system

Risk of injury due to unexpected motor start

If the vehicle's start/stop system is activated, the engine can start unexpectedly. A message in the dash panel insert indicates whether the start/stop system is activated.

- Deactivate start/stop system by switching off the ignition.

1.4 Safety precautions when using testers and measuring instruments during a road test

Risk of injury caused by unsecured testing and measuring instruments

When the front passenger airbag is triggered in an accident, insufficiently secured testing and measuring instruments become dangerous projectiles.

- Secure testing and measuring instruments on the rear seat.
- or
- Have a second person operate the test and measuring equipment on the rear seat.

1.5 Safety precautions when working on the cooling system

Danger of scalding by hot coolant

On a warm engine, the cooling system is under high pressure. Danger of scalding by steam and hot coolant.

- Wear protective gloves.
- Wear protective goggles.
- Reduce excess pressure by covering cap of coolant expansion tank with cloths and opening it carefully.

1.6 Safety precautions when working on vehicles with auxiliary/supplementary heater

Danger of fire and explosion from auxiliary/supplementary heater

In fire and explosion-endangered areas, a spark or the high temperature of the auxiliary/supplementary heater can cause a fire or an explosion. Risk of burns.

- Switch the auxiliary/supplementary heater off in fire and explosion-endangered areas.



Danger of poisoning from exhaust fumes

auxiliary/supplementary heaters produce poisonous exhaust fumes. Risk of poisoning and injuries to respiratory system.

- In closed areas, only switch on the auxiliary/supplementary heater when it is exclusively connected to an exhaust extractor system.
- In closed areas without an exhaust extractor system, switch off the auxiliary/supplementary heater.

Risk of damage when starting the engine

If components of the fuel system or coolant circuit of the auxiliary/supplementary heater are removed or opened, there is a possibility of causing damage to the auxiliary/supplementary heater.

- Never attempt to start the engine when components are removed or opened.

Malfunction caused by air in the fuel supply system

Upon repair work on the fuel tank or the fuel delivery unit, air will be drawn in by the metering pump and conveyed to the auxiliary/supplementary heater. Air in the fuel supply system may cause malfunction of the auxiliary/supplementary heater.

- Fill fuel take-off pipe with fuel.

Risk of accidents and risk of injury due to activated timer of auxiliary/supplementary heater

If the vehicle's timer for the auxiliary/supplementary heater is active, the heater may switch on unexpectedly. This poses a risk of poisoning from exhaust gases, a risk of burns caused by hot auxiliary/supplementary heater components as well as a risk of fire and explosions caused by high temperatures.

- Deactivate timer for auxiliary/supplementary heater.

1.7 Safety precautions when working on SCR system

Risk of injury from reducing agent

The reducing agent may cause eye and skin irritations as well as injuries to the respiratory system and poisoning.

- Wear protective goggles.
- Wear protective gloves.
- Always wear protective clothes.
- Make sure that a sufficient amount of fresh air is supplied. In closed areas, switch on the exhaust gas extractor system.



Risk of damage caused by reducing agent

If reducing agent gets onto trim panels and body components, the reducing agent may solidify after a while and thus damage the surface.

- Make sure that no reducing agent gets onto the trim panels or body components.
- If any reducing agent does get onto surfaces, remove it immediately using water and a cloth.

Risk of injury from reducing agent

The reducing agent may cause eye and skin irritations as well as injuries to the respiratory system and poisoning.

- Wear protective goggles.
- Wear protective gloves.
- Always wear protective clothes.
- Make sure that a sufficient amount of fresh air is supplied. In closed areas, switch on the exhaust gas extractor system.

Risk of damage caused by reducing agent

If reducing agent gets onto trim panels and body components, the reducing agent may solidify after a while and thus damage the surface.

- Make sure that no reducing agent gets onto the trim panels or body components.
- If any reducing agent does get onto surfaces, remove it immediately using water and a cloth.

When removing and installing components of the SCR system, note the following:

- ◆ The reducing agent tank must be empty when working on the SCR system. Information on when the reducing agent tank has to be emptied can be obtained from the respective description of work. Drain reducing agent tank ⇒ Rep. gr. 26 ; SCR system (selective catalytic reduction); Draining reducing agent tank .

Automatic return of reducing agent

- After the ignition has been switched off, the reducing agent is returned from the metering line leading to the reducing agent injector - N474- back to the reducing agent tank.
- Before any work is done in this area, it is necessary to wait until all the reducing agent has been returned to the tank; this can take up to 10 minutes after the ignition is switched off.
- Similarly, it is not permissible to disconnect the battery until all the reducing agent has been returned to the tank, i.e. 10 minutes after the ignition is switched off ⇒ Electrical system; Rep. gr. 27 ; Battery; Disconnecting and connecting battery .



2 General information

⇒ [“2.1 Type plates”, page 6](#)

⇒ [“2.2 Starting conditions for auxiliary/supplementary heater”, page 6](#)

⇒ [“2.3 Rules for cleanliness when working on auxiliary/supplementary heater and fuel system”, page 7](#)

2.1 Type plates

There are different types of type plates.

When the auxiliary heater is renewed, check the type plate and the duplicate plate and replace as necessary.

Type plate on auxiliary/supplementary heaters

- ◆ The date of first use is entered on the lower part of manufacturer's plate.
- ◆ The year the unit was first put into use must be entered on the “Genuine part” manufacturer's plate.

Webasto Therma & Comfort SE		E₁	
Made in Germany	CE		
Model Type	Therma Top Evo		
Operating Voltage			
Rated Output			
Fuel Type			
Working Pressure			
Part No.	Serial-No.		
Year of Manufacture			
			N00-11041

Duplicate type plate for auxiliary/supplementary heaters

- ◆ In the engine compartment of the vehicle, there is a second (duplicate) type plate attached to the right of the lock carrier (when looking in direction of travel). If there is any doubt concerning the type of the auxiliary/supplementary heater, always observe the type plate attached directly to the auxiliary/supplementary heater.

Factory label duplicate valid only with the original			
Webasto Therma & Comfort SE		E₁	
Made in Germany	CE		
Model Type	Therma Top Evo		
Operating Voltage / Rated Output			
Fuel Type			
Working Pressure			
Part No.			
			N00-11040

2.2 Starting conditions for auxiliary/supplementary heater

⇒ [“2.2.1 Starting conditions for auxiliary/supplementary heater, auxiliary heater mode”, page 6](#)

⇒ [“2.2.2 Starting conditions for auxiliary/supplementary heater, supplementary heater mode”, page 7](#)

2.2.1 Starting conditions for auxiliary/supplementary heater, auxiliary heater mode

- ◆ Auxiliary/supplementary heater coded in Gateway
- ◆ Reserve fuel level not reached
- ◆ No crash shut-off
- ◆ No event memory entries blocking start procedure
- ◆ Line filling carried out
- ◆ Transport mode deactivated



Note

The **A/C** button has no influence on the auxiliary heater operation.

2.2.2 Starting conditions for auxiliary/supplementary heater, supplementary heater mode

- ◆ Terminal 15 (ignition) on
- ◆ Engine speed > 600 rpm
- ◆ Coolant temperature < 72°C
- ◆ Ambient temperature < 5°C
- ◆ Reserve fuel level not reached
- ◆ On-board network manager not active
- ◆ No crash shut-off
- ◆ No event memory entries blocking start procedure
- ◆ Line filling carried out
- ◆ Supplementary heating function activated in multifunction display (MFD)



Note

The **A/C** button has no influence on the supplementary heater mode.

2.3 Rules for cleanliness when working on auxiliary/supplementary heater and fuel system

- ◆ Thoroughly clean all joints and surrounding areas before dismantling.
- ◆ Place removed parts on a clean surface and cover them over. Use lint-free cloths only.
- ◆ Carefully cover opened components or seal them if repairs cannot be carried out immediately.
- ◆ Install only clean parts; do not remove new parts from packaging until immediately before installing. Do not use any parts which have not been stored in their packaging (e.g. in a tool box).

If the fuel system is open:

- ◆ Do not work with compressed air
- ◆ Do not move the vehicle
- ◆ Do not start the engine
- ◆ Do not switch on auxiliary heater



3 Repair notes

⇒ [“3.1 Rules for cleanliness”, page 8](#)

⇒ [“3.2 General information”, page 8](#)

⇒ [“3.3 General repair instructions”, page 8](#)

⇒ [“3.4 Contact corrosion”, page 9](#)

⇒ [“3.5 Nuts and bolts”, page 9](#)

⇒ [“3.6 Routing and attachment of lines”, page 9](#)

3.1 Rules for cleanliness

Even small amounts of contamination/soiling can lead to defects. Therefore, observe the following rules for cleanliness when working on the air conditioning system:

- ◆ Seal open pipes and connections immediately with clean plugs, for example from the engine bung set - VAS 6122- .
- ◆ Place removed parts on a clean surface and cover them over. Use only lint-free cloths.
- ◆ If repair work cannot be performed immediately, carefully cover or seal components.
- ◆ Install only clean parts; do not remove new parts from packaging until immediately before installing. Do not use parts that have been stored outside their packaging (e. g. in tool boxes).
- ◆ Do not work with compressed air when the system is open.
- ◆ Protect disconnected electrical connectors from dirt and water, and only reconnect them in dry condition.

3.2 General information

- ◆ The engine control unit has a self-diagnosis capability. Before carrying out repairs and for fault finding, first read event memory.
- ◆ For trouble-free operation of electrical components, a voltage of at least 11.5 volts is necessary.
- ◆ Do not use sealants containing silicone. Particles of silicone drawn into the engine will not be burnt in the engine and damage the Lambda probe.
- ◆ Vehicles are fitted with a crash fuel shut-off circuit. It reduces the danger of a fire in a crash as the fuel pump is switched off via the fuel pump relay.
- ◆ The system also improves the starting characteristics of the engine. When the driver door is opened, the fuel pump is activated for 2 seconds to build up pressure in the fuel system.

3.3 General repair instructions

Never attempt to repair any components of the auxiliary/supplementary heater. Defective components must be replaced with Genuine parts.

- Read self-diagnosis of auxiliary/supplementary heater using
⇒ Vehicle diagnostic tester.
- When parts of the fuel system are removed or renewed, ensure that all components for fuel delivery to the auxiliary/supplementary heater are properly installed.
- Mark coolant hoses before removing to avoid any confusion when reconnecting.



Following repair work in the vicinity of the auxiliary/supplementary heater fuel line, check the following points:

- ◆ that the fuel lines are routed flush with the vehicle underbody and are protected against mechanical damage.
- ◆ that the auxiliary/supplementary heater fuel line is protected from heating which might disturb operation.
- ◆ The fuel line does not come in contact with parts that become hot.

3.4 Contact corrosion

Contact corrosion can occur if unsuitable connecting elements (screws, bolts, nuts, washers, etc.) are used.

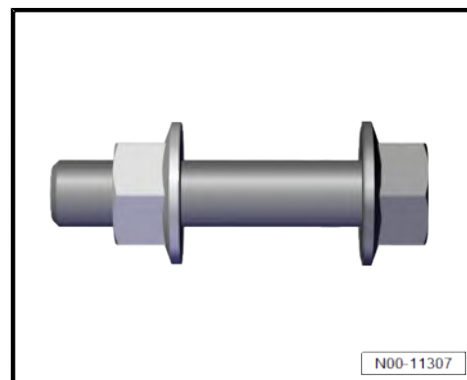
For this reason, only connecting elements with a special surface coating have been fitted.

In addition, rubber, plastic and adhesives are made of non-conductive materials.

If there is any doubt about the suitability of parts, a general rule is to use new parts ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

3.5 Nuts and bolts

- ◆ The provided specified torques apply for uncoiled nuts, bolts and screws.
- ◆ Always renew self-locking nuts and bolts.
- ◆ Always renew nuts and bolts with turning further angle.



3.6 Routing and attachment of lines

Risk of damage to lines

Lines may become damaged by moving or hot components.

- Route lines in their original positions.
- Make sure there is sufficient clearance between the lines and moving as well as hot components.



82 – Supplementary heating

1 Overview of fitting locations - auxiliary/supplementary heater

⇒ [“1.1 Overview of fitting locations - components not located in passenger compartment”, page 10](#)

⇒ [“1.2 Overview of fitting locations - auxiliary air heater located in passenger compartment”, page 11](#)

1.1 Overview of fitting locations - components not located in passenger compartment

1 - Temperature sensor on roof for air inlet on auxiliary air heater - G967-

- ❑ Removing and installing ⇒ [page 71](#)

2 - Roof aerial

- ❑ Removing and installing ⇒ Communication; Rep. gr. 91 ; Aerial systems; Removing and installing roof aerial

3 - Remote control receiver for auxiliary coolant heater - R149-

- ❑ Removing and installing ⇒ [page 68](#)

4 - Ambient temperature sensor - G17-

- ❑ Removing and installing ⇒ [page 68](#)

5 - Shut-off valve

- ❑ Removing and installing ⇒ [page 50](#)

6 - Auxiliary/supplementary heater

- ❑ Removing and installing ⇒ [page 17](#)

7 - Fuel pump

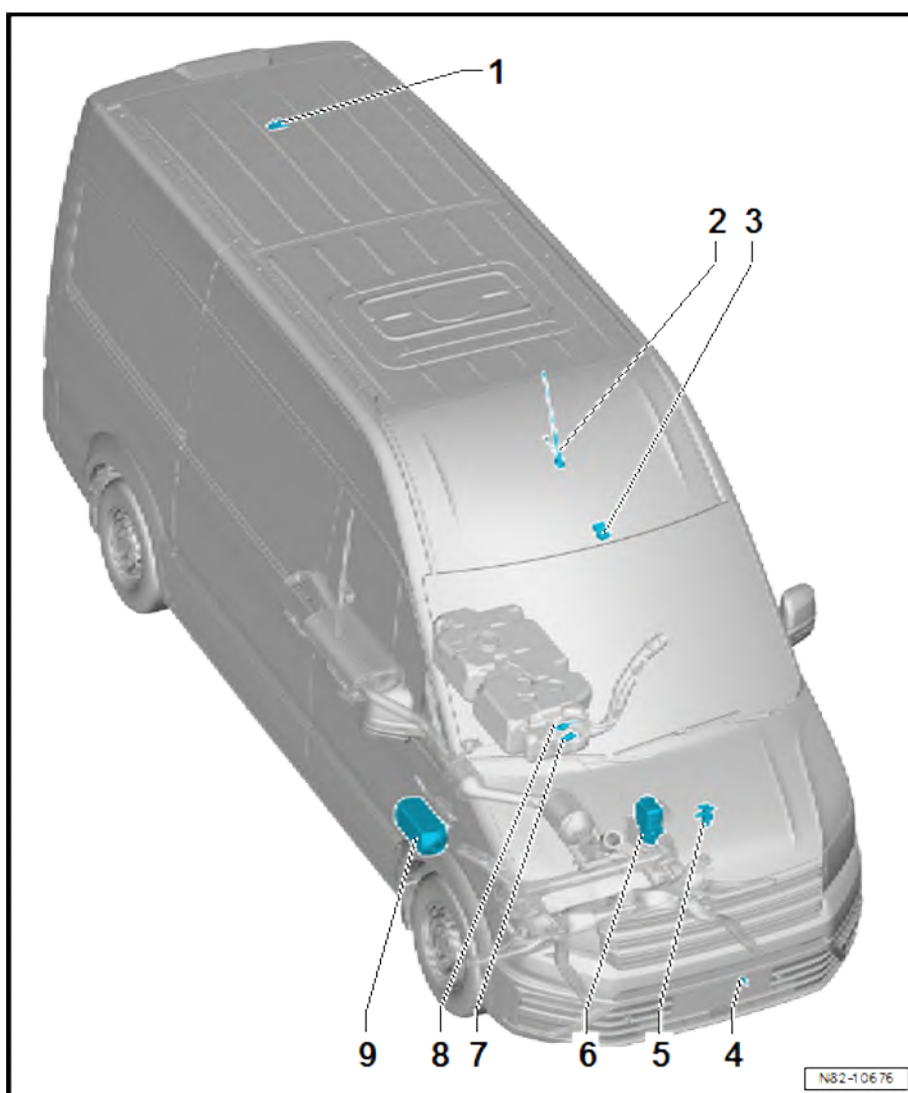
- ❑ For auxiliary air heater
- ❑ Removing and installing ⇒ [page 56](#)

8 - Metering pump - V54-

- ❑ For auxiliary coolant heater
- ❑ Removing and installing ⇒ [page 55](#)

9 - Auxiliary air heater

- ❑ Removing and installing ⇒ [page 19](#)
- ❑ Dismantling and assembling ⇒ [page 21](#)





1.2 Overview of fitting locations - auxiliary air heater located in passenger compartment

1 - Operating and display unit in roof console

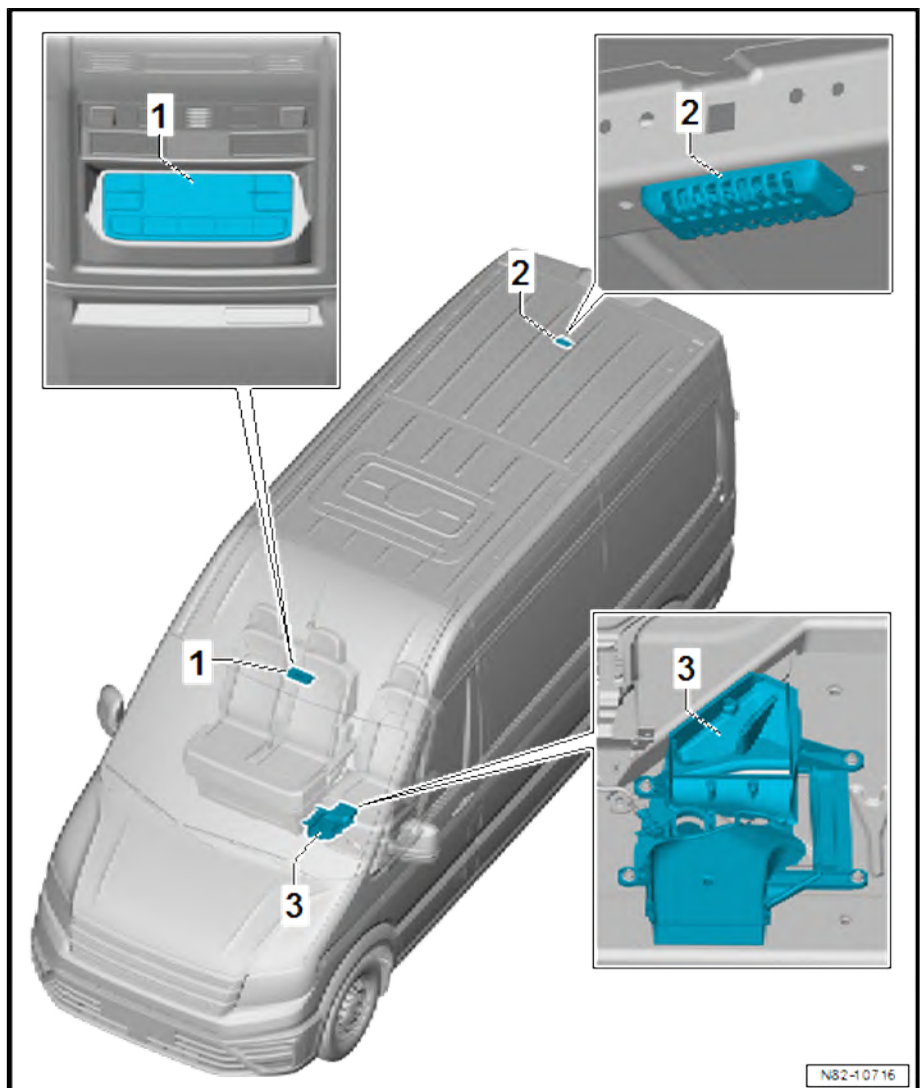
- ❑ Removing and installing
⇒ Heater and fresh air ventilation system; Rep. gr. 87 ; Operating and display unit; Removing and installing operating and display unit

2 - Temperature sensor on roof for air inlet on auxiliary air heater - G967-

- ❑ Removing and installing
⇒ [page 71](#)

3 - Distribution box, auxiliary air heater

- ❑ Removing and installing
⇒ [page 41](#)





2 Auxiliary/supplementary heater

⇒ [“2.1 Overview of fitting locations - auxiliary/supplementary heater”, page 12](#)

⇒ [“2.2 Assembly overview - auxiliary heater”, page 14](#)

⇒ [“2.3 Removing and installing auxiliary/supplementary heater”, page 17](#)

⇒ [“2.4 Dismantling and assembling auxiliary/supplementary heater”, page 21](#)

⇒ [“2.5 Removing and installing air intake silencer”, page 29](#)

⇒ [“2.6 Removing and installing circulation pump V55 ”, page 30](#)

⇒ [“2.7 Removing and installing exhaust system”, page 32](#)

⇒ [“2.8 Removing and installing auxiliary/supplementary heater bracket”, page 33](#)

⇒ [“2.9 Removing and installing cover”, page 37](#)

⇒ [“2.10 Dismantling and assembling heater unit”, page 38](#)

⇒ [“2.11 Removing and installing glow plug with flame monitor Q8 ”, page 39](#)

⇒ [“2.12 Removing and installing temperature sensor G18 / overheating sensor G189 ”, page 40](#)

⇒ [“2.13 Removing and installing auxiliary heater control unit J364 ”, page 41](#)

⇒ [“2.14 Removing and installing combustion air blower V6 ”, page 41](#)

⇒ [“2.15 Removing and installing distribution box, auxiliary air heater”, page 41](#)

⇒ [“2.16 Removing and installing bottom vents, auxiliary air heater”, page 43](#)

⇒ [“2.17 Removing and installing vent duct for auxiliary air heater”, page 43](#)

⇒ [“2.18 Removing and installing intake port for auxiliary air heater”, page 44](#)

⇒ [“2.19 Removing and installing stone deflector for auxiliary air heater”, page 46](#)

⇒ [“2.20 Removing and installing bracket for vent duct of auxiliary air heater”, page 46](#)

2.1 Overview of fitting locations - auxiliary/supplementary heater

Overview of fitting locations - auxiliary heater, auxiliary coolant heater ⇒ [page 12](#)

Overview of fitting locations - auxiliary heater, auxiliary air heater ⇒ [page 13](#)

Overview of fitting locations - auxiliary heater, auxiliary coolant heater



1 - 3/2 way valve

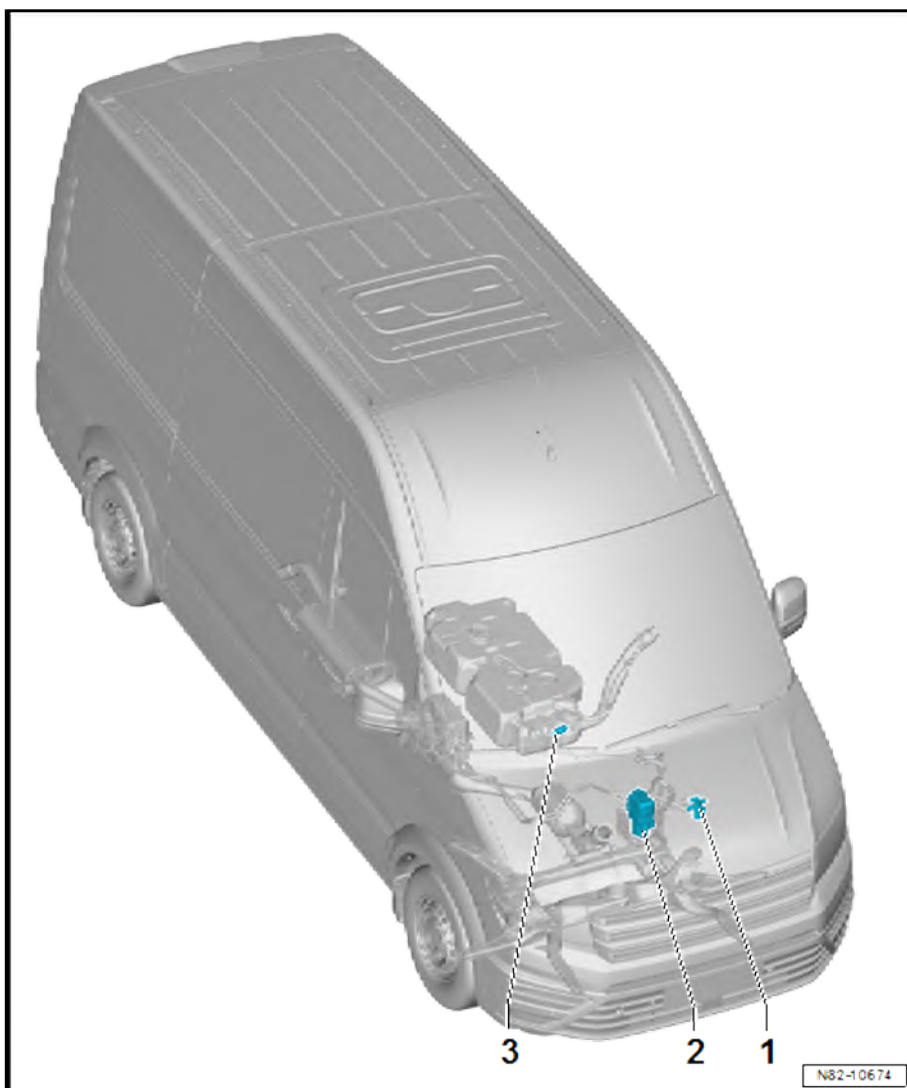
- ☐ Removing and installing
⇒ [page 50](#)

2 - Auxiliary/supplementary heater

- ☐ Removing and installing
⇒ [page 17](#)

3 - Metering pump - V54-

- ☐ For auxiliary coolant heater
- ☐ Removing and installing
⇒ [page 55](#)



Overview of fitting locations - auxiliary heater, auxiliary air heater



1 - Fuel pump

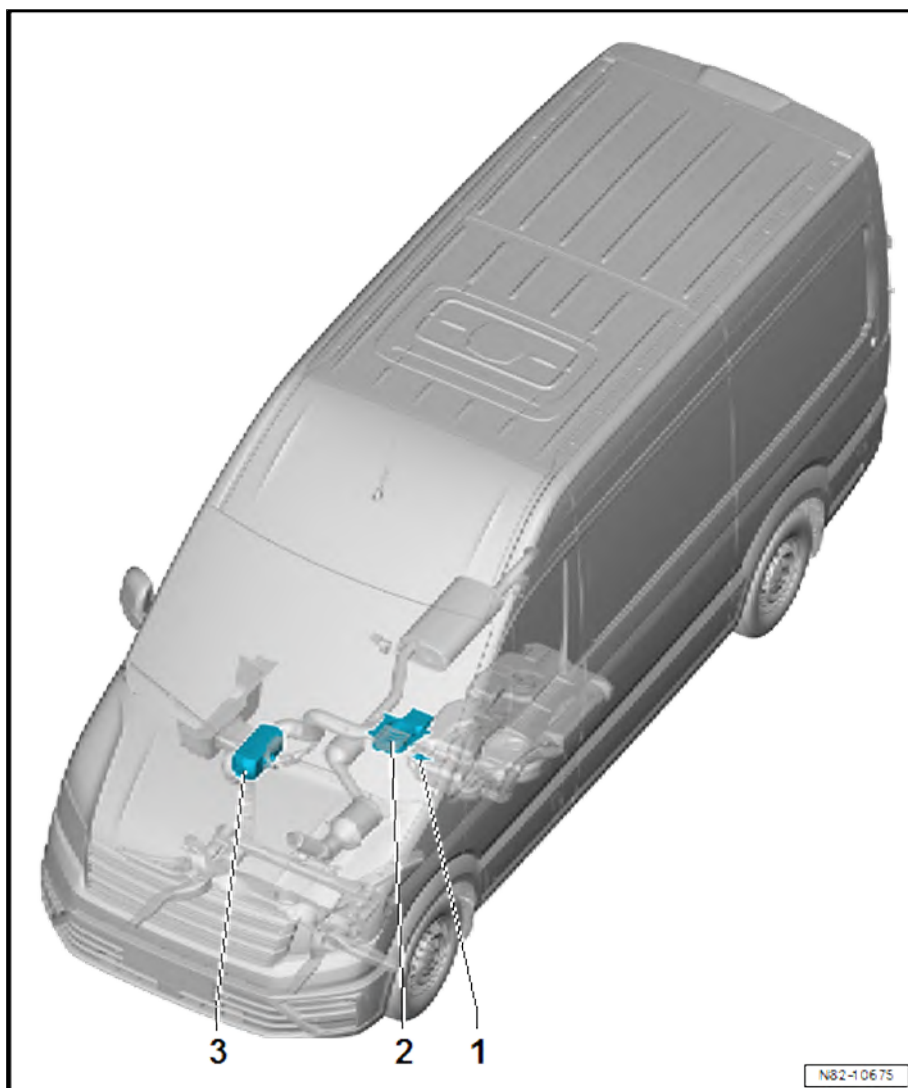
- ☐ For auxiliary air heater
- ☐ Removing and installing
⇒ [page 55](#)

2 - Distribution box

- ☐ With vent
- ☐ Removing and installing
⇒ [page 41](#)
- ☐ Removing and installing
air distribution flap control motor 2 - V667-
⇒ [page 72](#)

3 - Auxiliary air heater

- ☐ Assembly overview
⇒ [page 14](#)



2.2 Assembly overview - auxiliary heater

Assembly overview - auxiliary heater, auxiliary coolant heater
⇒ [page 14](#)

Assembly overview - auxiliary heater, auxiliary air heater
⇒ [page 15](#)

Assembly overview - auxiliary heater, auxiliary coolant heater



1 - Bracket

- ☐ Removing and installing
⇒ [page 33](#)

2 - Bearings

3 - Bracket

4 - Bolt

- ☐ 8 Nm

5 - Bolts

- ☐ Qty. 4
- ☐ 6 Nm

6 - Bracket

7 - Intake manifold

8 - Air intake silencer

- ☐ Removing and installing
⇒ [page 29](#)

9 - Retaining rubber

10 - Circulation pump - V55-

- ☐ Removing and installing
⇒ [page 30](#)

11 - Spring-type clip

- ☐ Qty. 4

12 - Coolant hose

13 - Thermostat

14 - Coolant hose

15 - Exhaust system

- ☐ Removing and installing
⇒ [page 32](#)

16 - Auxiliary/supplementary heater

- ☐ Removing and installing ⇒ [page 17](#)

17 - Fuel line to metering pump - V54-

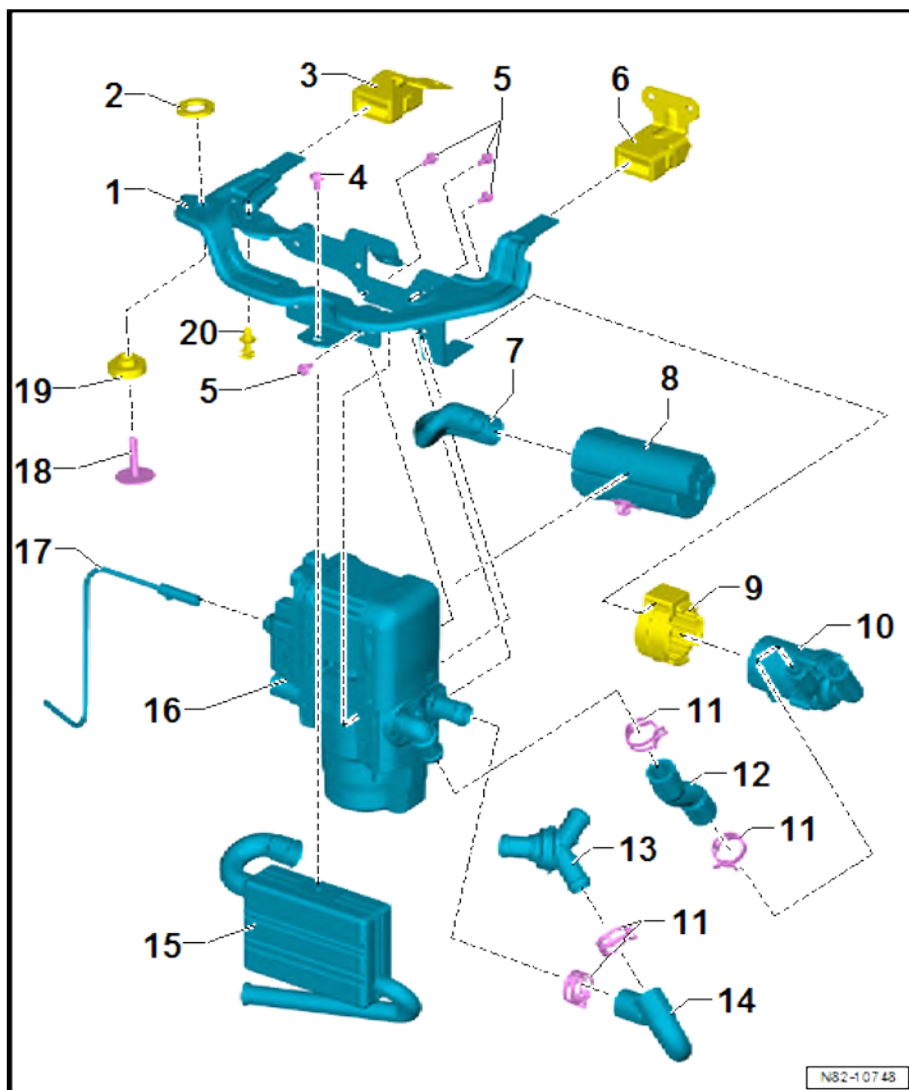
- ☐ Removing and installing ⇒ [page 57](#)

18 - Bolt

- ☐ 20 Nm

19 - Bearings

20 - Clip



Assembly overview - auxiliary heater, auxiliary air heater



1 - Auxiliary air heater

- ☐ Removing and installing
⇒ [page 19](#)
- ☐ Dismantling and assembling
⇒ [page 21](#)

2 - Vent duct

- ☐ Removing and installing
⇒ [page 43](#)

3 - Bracket

- ☐ For vent duct, auxiliary air heater
- ☐ Removing and installing
⇒ [page 46](#)

4 - Nuts

- ☐ Qty. 2
- ☐ 8 Nm

5 - Air intake silencer

- ☐ Removing and installing
⇒ [page 30](#)

6 - Bracket for auxiliary air heater

- ☐ On heater unit
- ☐ Removing and installing
⇒ [page 35](#)

7 - Nuts

- ☐ Qty. 4
- ☐ 8 Nm

8 - Bearings

- ☐ With bush

9 - Bolt

- ☐ 20 Nm

10 - Screw-type clips

- ☐ Qty. 2
- ☐ 6 Nm

11 - Exhaust system

- ☐ With silencer
- ☐ Removing and installing ⇒ [page 32](#)

12 - Bolt

- ☐ 8 Nm

13 - Tube

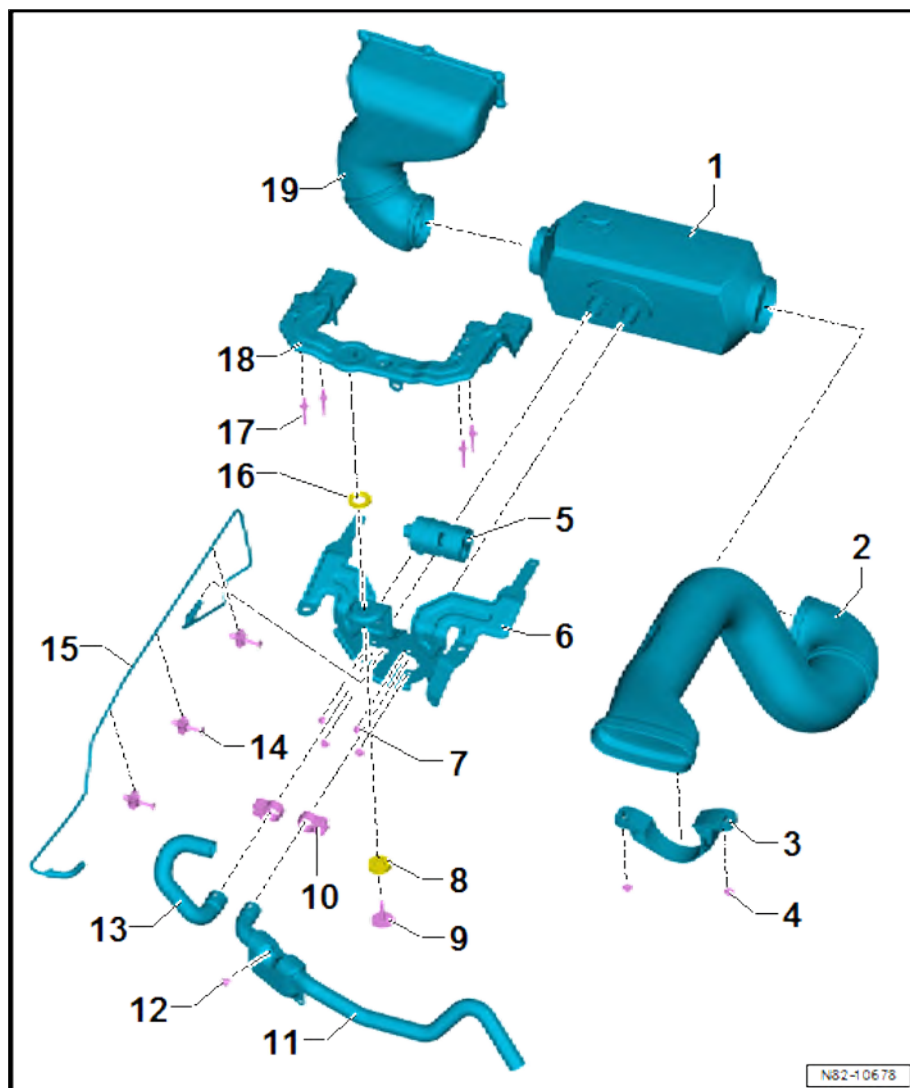
- ☐ Intake air inlet
- ☐ Removing and installing ⇒ [page 30](#)

14 - Wiring retainer

- ☐ Qty. 5
- ☐ For fuel line to fuel pump - heater unit

15 - Fuel line

- ☐ To fuel pump
- ☐ Removing and installing ⇒ [page 59](#)





16 - Bearings

17 - Pop rivets

- ☐ Qty. 4
- ☐ Bracket for on body

18 - Bracket for auxiliary air heater

- ☐ On body
- ☐ Removing and installing ⇒ [page 35](#)

19 - Intake duct

- ☐ Removing and installing ⇒ [page 44](#)

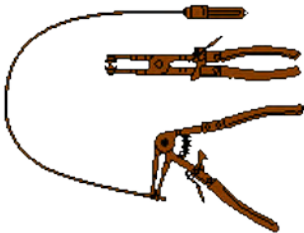
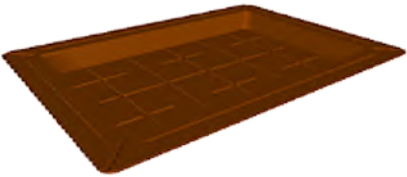
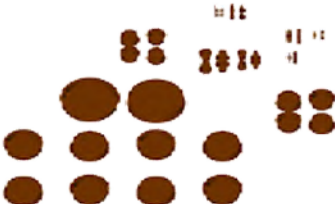
2.3 Removing and installing auxiliary/supplementary heater

⇒ [“2.3.1 Removing and installing auxiliary heater, auxiliary coolant heater”, page 17](#)

⇒ [“2.3.2 Removing and installing auxiliary heater, auxiliary air heater”, page 19](#)

2.3.1 Removing and installing auxiliary heater, auxiliary coolant heater

Special tools and workshop equipment required

V.A.G 1410 	3094 
VA S 5024 A 	VA S 6208 
VA S 6122 	



- ◆ Torque wrench - V.A.G 1410-
- ◆ Hose clamps to 25 mm - 3094-
- ◆ Spring-type clip pliers - VAS 5024A- or hose clip pliers - VAS 6340-
- ◆ Drip tray for workshop hoist - VAS 6208-
- ◆ Set of plugs for engine - VAS 6122-
- ◆ Vehicle diagnostic tester

Removing



Note

Connect the ⇒ Vehicle diagnostic tester before renewing the auxiliary heater.

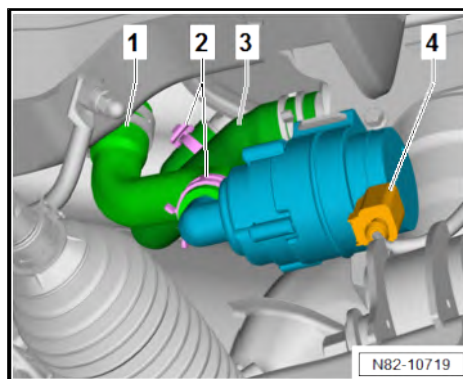
- Disconnect battery ⇒ Electrical system; Rep. gr. 27 ; Battery; Disconnecting and reconnecting battery .
- Remove front left underbody cladding ⇒ General body repairs, exterior; Rep. gr. 66 ; Underbody cladding; Removing and installing underbody cladding .
- Remove front centre heat shield ⇒ General body repairs, exterior; Rep. gr. 66 ; Mouldings/trims/extensions; Removing and installing floor heat shield .
- If fitted, remove noise insulation ⇒ General body repairs, exterior; Rep. gr. 66 ; Noise insulation; Removing and installing noise insulation .

CAUTION

On a warm engine, the cooling system is under high pressure. Danger of scalding by steam and hot coolant.

Skin and other parts of the body may be scalded.

- Wear protective gloves.
 - Wear protective goggles.
 - Reduce excess pressure by covering cap of coolant expansion tank with cloths and opening it carefully.
-
- Clamp off coolant hoses -1- and -3- using hose clamps for hoses up to 25 mm - 3094- .
 - Open spring-type clips -2- of coolant hoses using hose clip pliers - VAS 5024A- or hose clips pliers - VAS 6340- .
 - Place drip tray for workshop hoist - VAS 6208- underneath.
 - Pull off coolant hoses -1- and -3-.
 - Disconnect electrical connector -4-.





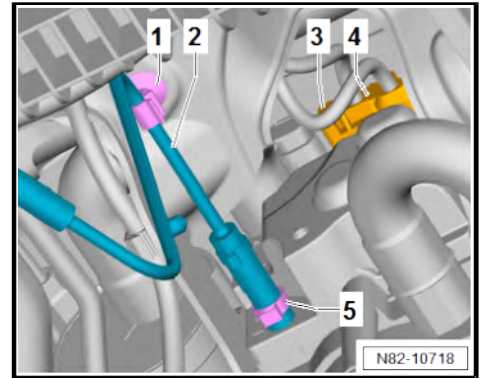
- Unclip fuel line -2- from retainer -1-.
- Loosen clamp -5-.

⚠ CAUTION

The fuel system is pressurised.

Danger of injury caused by fuel spray.

- Wear protective goggles.
- Wear protective gloves.
- To release pressure, wrap a clean cloth around the connection and carefully loosen the connection.



- Disconnect fuel line -2-.
- Seal fuel line -2- with a bung from engine bung set - VAS 6122- .
- Disconnect electrical connectors -3- and -4-.
- Unscrew bolt -2- from bracket -1-.
- Pull auxiliary heater -1- out of rubber mountings -3- and -4-.
- Remove auxiliary heater -1-.

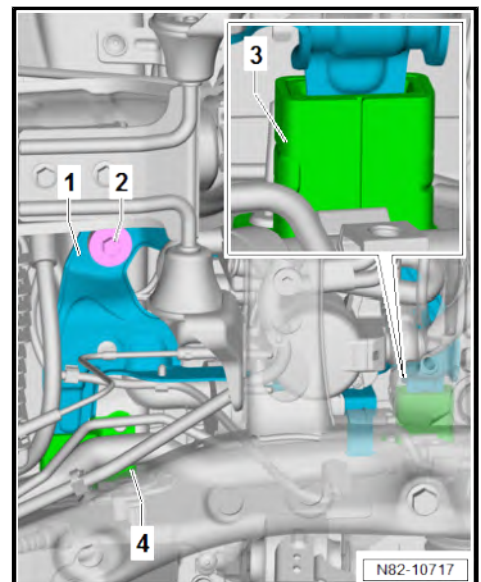
Installing

Install in reverse order of removal, observing the following:

- Fill up with coolant ⇒ Rep. gr. 19 ; Cooling system/coolant; Draining and filling coolant .
- Bleed cooling system ⇒ Rep. gr. 19 ; Cooling system, coolant; Draining and filling with coolant .
- Perform functional check of auxiliary heater.

Specified torques

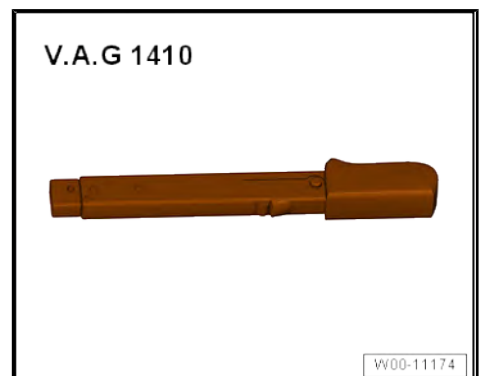
- ◆ ⇒ ["2.2 Assembly overview - auxiliary heater", page 14](#)



2.3.2 Removing and installing auxiliary heater, auxiliary air heater

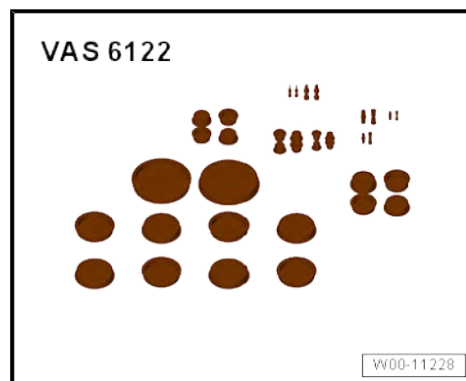
Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1410-





- ◆ Set of plugs for engine - VAS 6122-



Removing

- Disconnect battery ⇒ Electrical system; Rep. gr. 27 ; Battery; Disconnecting and reconnecting battery .
- Remove front right underbody cladding ⇒ General body repairs, exterior; Rep. gr. 66 ; Underbody cladding; Removing and installing underbody cladding .

Vehicles with PR number 2D0

- Remove stone deflector ⇒ [page 46](#) .

Continued for all vehicles

CAUTION

Danger of severe burns from hot exhaust system.

Burns to hands and other body parts possible.

- Allow exhaust system to cool.

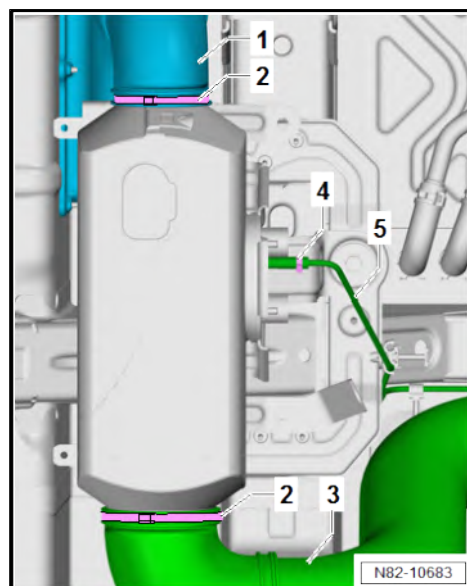
- Remove cable tie -2-.
- Pull off intake duct -1-.
- Pull off vent duct -3-.
- Remove clamp -4-.

CAUTION

The fuel system is pressurised.

Danger of injury caused by fuel spray.

- Wear protective goggles.
 - Wear protective gloves.
 - To release pressure, wrap a clean cloth around the connection and carefully loosen the connection.
- Disconnect fuel line -5-.
 - Seal fuel line -5- with clean plugs from engine bung set - VAS 6122- .



Note

For reasons of clarity, the exhaust system and air intake silencer are shown removed in the following illustration.



- Unscrew bolt -3-.
- Disconnect electrical connector -2-.
- Pull auxiliary air heater -1- together with bracket in direction of exhaust system out of damper element holder.

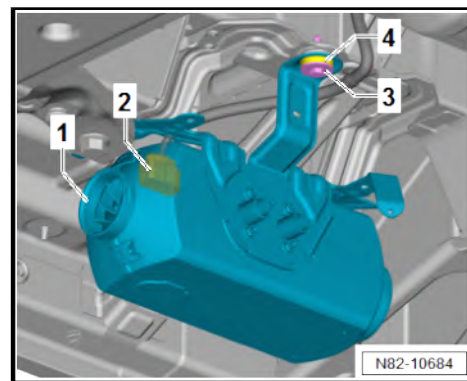
Installing

Install in reverse order of removal, observing the following:

- Inspect mountings -4- on bracket for damage, and renew them if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ web-MANTIS for MAN TGE.

Specified torques

- ◆ ⇒ [“2.2 Assembly overview - auxiliary heater”, page 14](#)



2.4 Dismantling and assembling auxiliary/supplementary heater

⇒ [“2.4.1 Removing and installing lower casing”, page 21](#)

⇒ [“2.4.2 Removing and installing auxiliary air heater control unit J604”, page 22](#)

⇒ [“2.4.3 Removing and installing combination sensor”, page 23](#)

⇒ [“2.4.4 Checking combination sensor”, page 24](#)

⇒ [“2.4.5 Removing and installing combustion air blower 2 V385”, page 26](#)

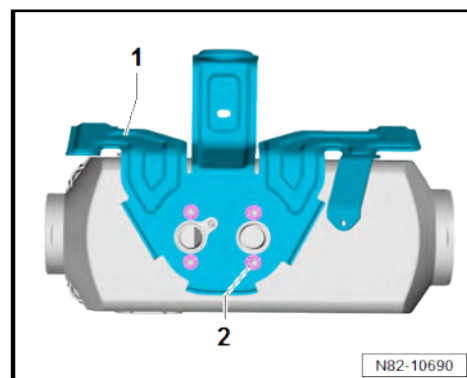
⇒ [“2.4.6 Removing and installing combustion chamber”, page 27](#)

⇒ [“2.4.7 Removing and installing glow plug for heater Q9”, page 29](#)

2.4.1 Removing and installing lower casing

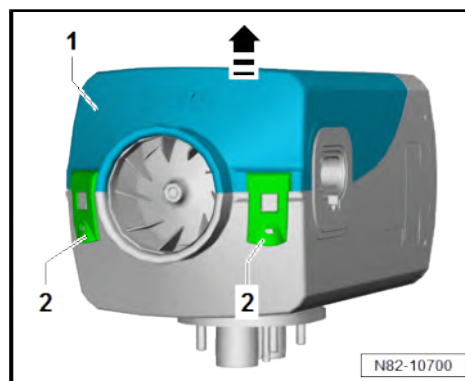
Removing

- Remove auxiliary air heater ⇒ [page 19](#) .
- Remove air intake silencer ⇒ [page 30](#) .
- Remove exhaust system ⇒ [page 32](#) .
- Unscrew nuts -2-.
- Remove bracket -1-.

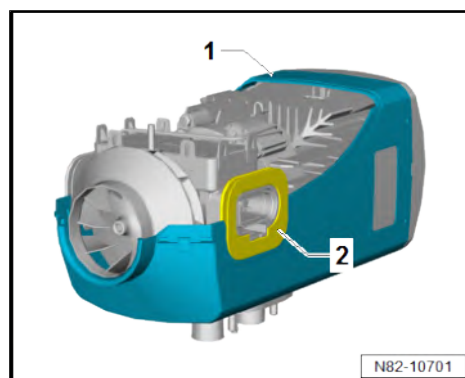




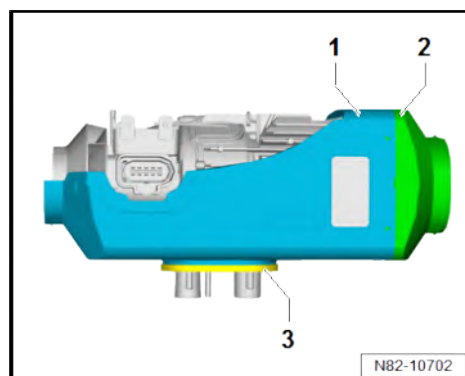
- Release locking tabs -2-.
- Swing cover -1- in direction of -arrow-, and detach it.



- Remove connector grommet -2- from lower casing -1-.



- Pull air outlet cowling -2- off lower casing -1-.
- Pull off flange seal -3-.



- Swing heater unit -2- in direction of -arrow- out of lower casing -1-, and remove it.

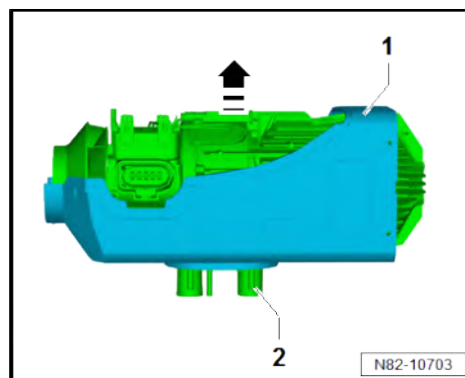
Installing

Install in reverse order of removal, observing the following:



Note

If the lower casing is to be renewed, attach a new type plate, and transfer the data from the old type plate to the new one.

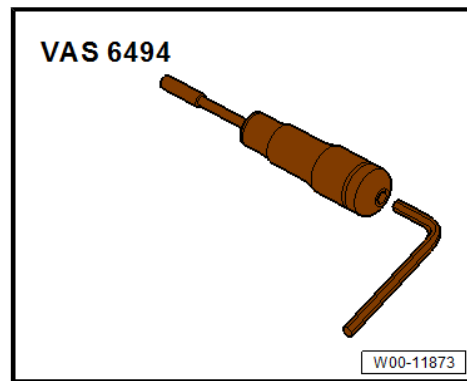


2.4.2 Removing and installing auxiliary air heater control unit - J604-

Special tools and workshop equipment required

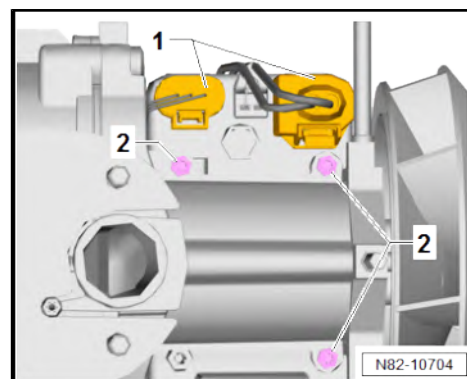


- ◆ Torque screwdriver - VAS 6494-



Removing

- Remove lower casing ➤ [page 21](#) .
- Separate electrical connectors -1-.
- Move wiring harnesses aside.
- Unscrew bolts -2-.



- Detach auxiliary air heater control unit - J604- -1- upwards from heater unit -2-.

Installing

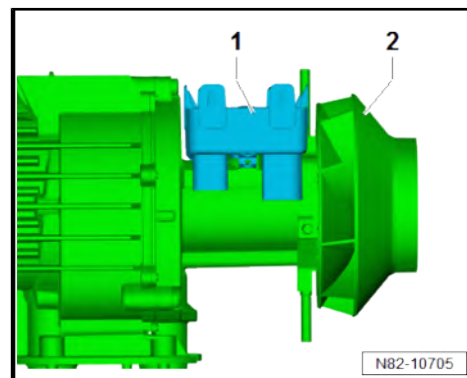
Install in reverse order of removal, observing the following:



Note

When installing, make sure that the 10-pin connection is located on the glow plug end.

- Screw in bolts -2- loosely.
- Tighten bolts -2- alternately in diagonal sequence to specified torque.



Specified torques

Component	Specified torque
Bolts -2-	1.5 Nm

2.4.3 Removing and installing combination sensor



Note

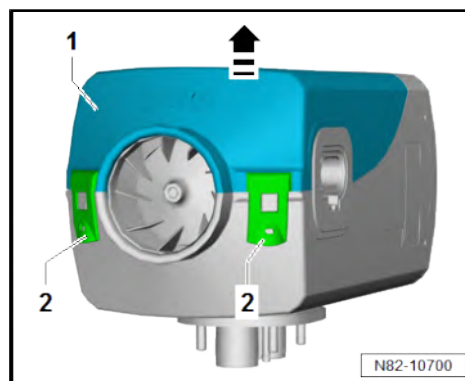
The combination sensor comprises the flame monitor - G64- and overheating sensor - G189- in one component.

Removing

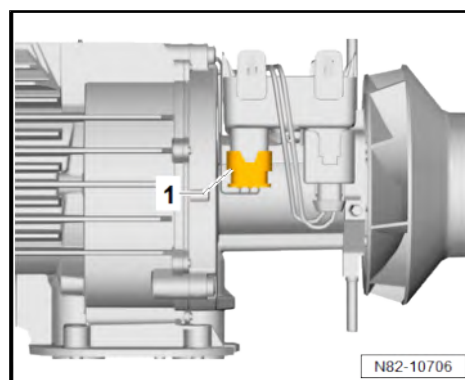
- Remove auxiliary air heater ➤ [page 19](#) .



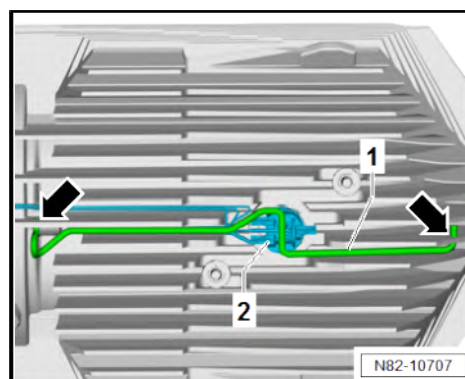
- Release locking tabs -2-.
- Swing cover -1- in direction of -arrow-, and detach it.



- Disconnect connector -1-.



- Release retaining clip -1- in area marked by -arrows-, and remove it.
- Pull combination sensor -2- out of heater unit.



Installing

Install in reverse order of removal, observing the following:

- Rout wiring harness through nearest cooling fin.



Note

Make sure not to damage the wiring harness when inserting the retaining clip.

2.4.4 Checking combination sensor

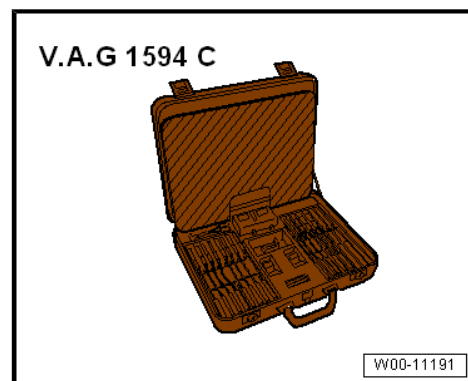
Special tools and workshop equipment required

- ◆ Hand multimeter - V.A.G 1526 E-





◆ Auxiliary measuring set - V.A.G 1594C-



Note

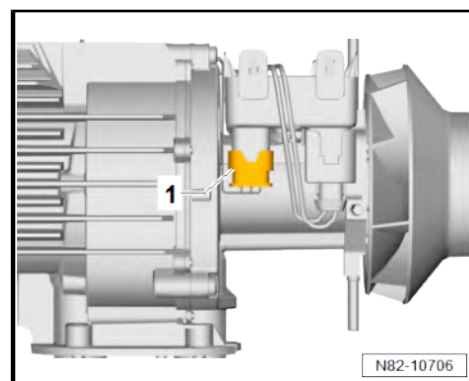
The combination sensor comprises the flame monitor - G64- and overheating sensor - G189- in one component.

Sequence of operations

- Remove lower casing ➔ [page 21](#) .
- Disconnect connector -1-.

Test procedure

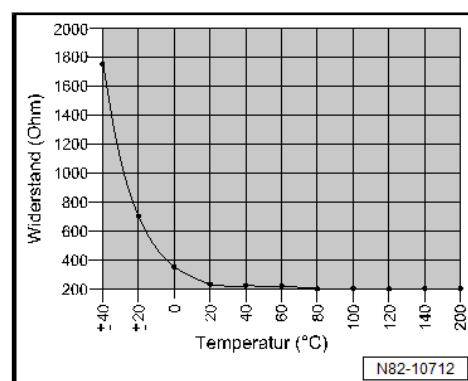
- Using hand-held multimeter - V.A.G 1526 E- and auxiliary measuring set - V.A.G 1594C- , perform resistance test observing the latest ➔ Current flow diagrams, Electrical fault finding and Fitting locations.



Note

Renew the combination sensor if the resistance value does not match the curve in diagram or the table of values ➔ Electronic parts catalogue (ETKA) or ➔ webMANTIS for MAN TGE.

Diagram and table of values for overheating sensor - G189-

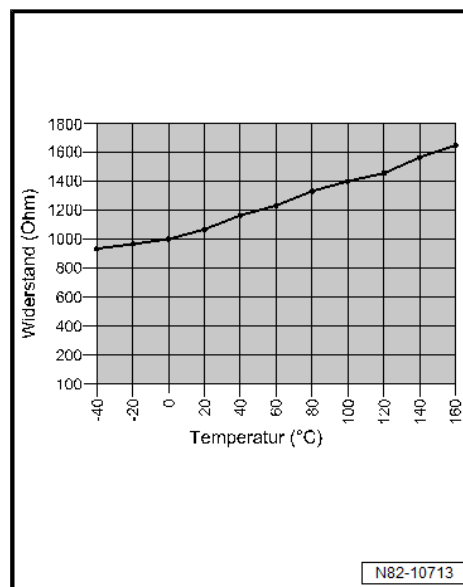


Temperature in °C	Min. resistance, kΩ	Max. resistance, kΩ
-40	1597	1913
-20	458.80	533.40
0	154.70	175.50
20	59.30	65.84
40	25.02	28.04
60	11.56	13.16
80	5.78	6.68



Temperature in °C	Min. resistance, kΩ	Max. resistance, kΩ
100	3.09	3.62
120	1.75	2.08
140	1.05	1.26
160	0.66	0.79
180	0.43	0.52
200	0.29	0.35

Diagram and table of values for flame monitor - G64-



Temperature in °C	Min. resistance, Ω	Max. resistance, Ω
-40	825.9	859.6
-20	803.2	940
0	980	1020
20	1056.4	1099.5
40	1132.3	1178.5
60	1207.8	1257.1
80	1282.8	1335.1
100	1357.4	1412.8
120	1431.5	1489.9
140	1505.1	1566.6
160	1578.3	1642.8

2.4.5 Removing and installing combustion air blower 2 - V385-

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1410-

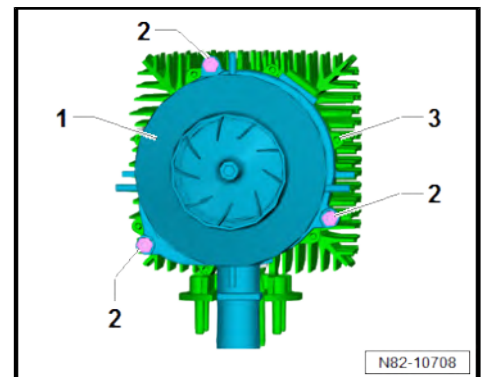
V.A.G 1410



WV00-11174

Removing

- Remove auxiliary air heater control unit - J604- ⇒ [page 22](#) .
- Unscrew bolts -2-.
- Detach combustion air blower 2 - V385- -1- from heat exchanger -3-.



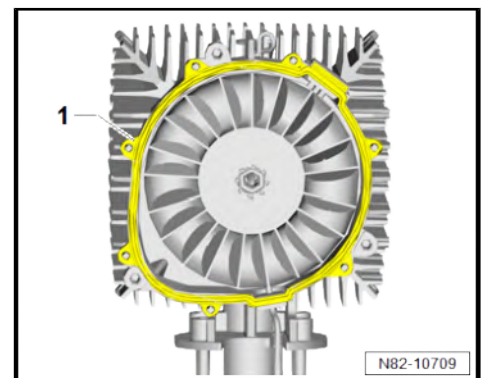
N82-10708

Installing

Install in reverse order of removal, observing the following:

- Renew seal -1-.

Specified torques



N82-10709

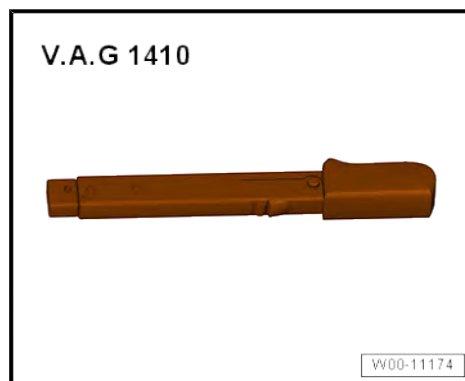
Component	Specified torque
Bolts -2-	6.5 Nm

2.4.6 Removing and installing combustion chamber

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1410-



Removing

Remove combustion air blower 2 - V385- ➔ [page 26](#) .

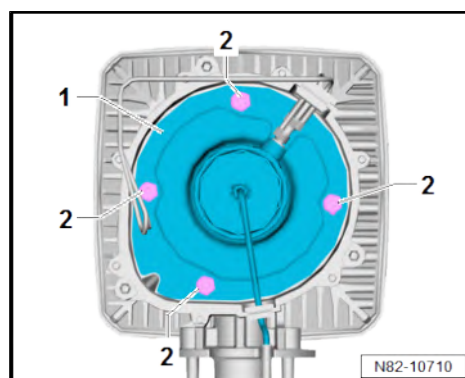
- Unscrew bolts -2-.



Note

Combustion chamber -1- is removed along with glow plug for heater - Q9- and fuel pipe.

- Pull out combustion chamber -1-.



Installing

Install in reverse order of removal, observing the following:

- Renew seal between combustion chamber -1- and heat exchanger.



Note

When reassembling, use the rubber grommets for the wiring harness of the glow plug for heater - Q9- and fuel pipe as a guide.

- Insert combustion chamber -1-.
- Screw in bolts -2- loosely.
- Tighten bolts -2- alternately in diagonal sequence to specified torque.

Specified torques

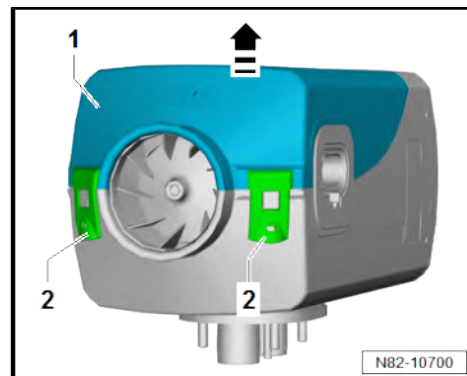
Component	Specified torque
Bolts -2-	6.5 Nm



2.4.7 Removing and installing glow plug for heater - Q9-

Removing

- Remove auxiliary air heater ⇒ [page 19](#) .
- Release locking tabs -1-.
- Swing cover -2- in direction of -arrow-, and detach it.



- Pull grommet -3- out of lower casing -1-.
- Disconnect electrical connector -2-.
- Lever out grommet -4- with a screwdriver.
- Remove glow plug for heater - Q9- .

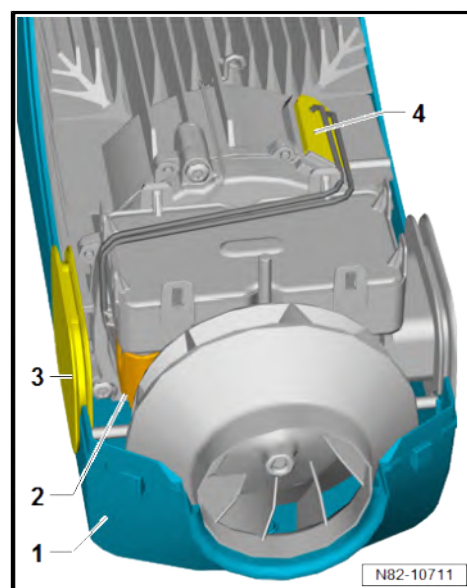
Installing

Install in reverse order of removal, observing the following:



Note

Make sure that the locking element of the electrical connector for the glow plug for heater - Q9- points towards the combustion air blower 2 - V385- .



2.5 Removing and installing air intake silencer

⇒ [“2.5.1 Removing and installing air intake silencer, auxiliary coolant heater”, page 29](#)

⇒ [“2.5.2 Removing and installing air intake silencer, auxiliary air heater”, page 30](#)

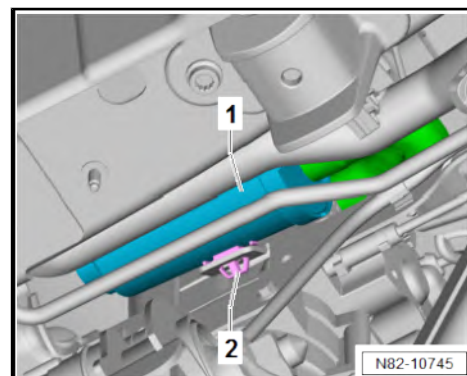
2.5.1 Removing and installing air intake silencer, auxiliary coolant heater

Removing

- Remove front left underbody cladding ⇒ General body repairs, exterior; Rep. gr. 66 ; Underbody cladding; Removing and installing underbody cladding .
- Press together spreader clip -2-.
- Detach air intake silencer -1- from auxiliary heater.

Installing

Install in reverse order of removal.

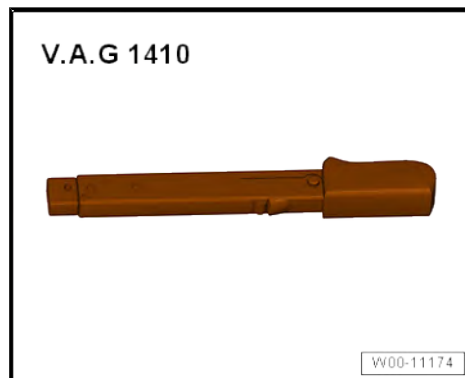




2.5.2 Removing and installing air intake silencer, auxiliary air heater

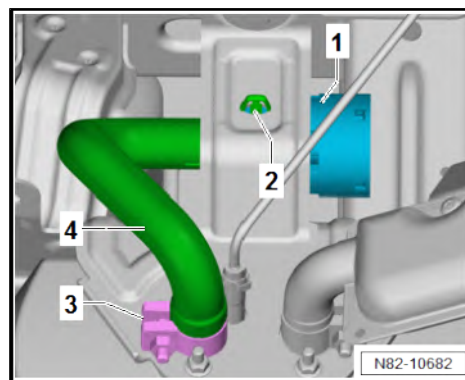
Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1410-



Removing

- If fitted, remove front right underbody cladding ⇒ General body repairs, exterior; Rep. gr. 66 ; Underbody cladding; Removing and installing underbody cladding .
- Loosen screw-type clip -3-.
- Press together locking lug -2-.
- Remove air intake silencer -1- together with pipe -4-.
- Turn air intake silencer -1- clockwise to detach it from pipe -4-.



Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“2.2 Assembly overview - auxiliary heater”, page 14](#)

2.6 Removing and installing circulation pump - V55-

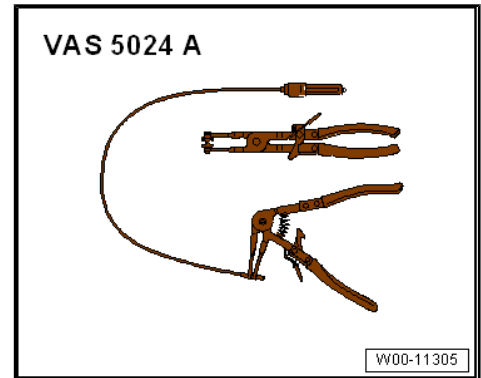
Special tools and workshop equipment required

- ◆ Hose clamps to 25 mm - 3094-





- ◆ Spring-type clip pliers - VAS 5024A- or hose clip pliers - VAS 6340-



- ◆ Drip tray for workshop hoist - VAS 6208-



Removing

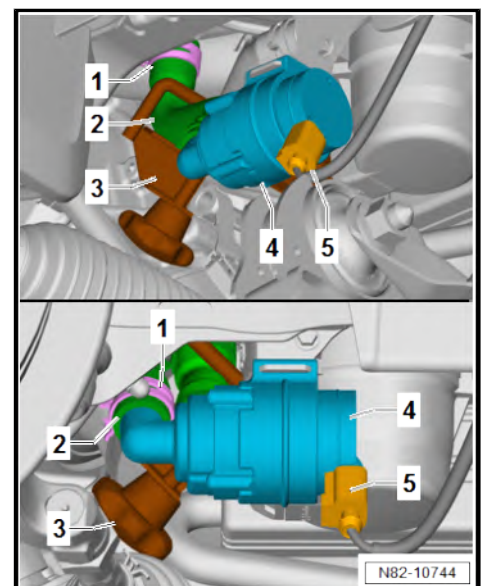
- Remove front left underbody cladding ⇒ General body repairs, exterior; Rep. gr. 66 ; Underbody cladding; Removing and installing underbody cladding .

CAUTION

On a warm engine, the cooling system is under high pressure. Danger of scalding by steam and hot coolant.

Skin and other parts of the body may be scalded.

- Wear protective gloves.
- Wear protective goggles.
- Reduce excess pressure by covering cap of coolant expansion tank with cloths and opening it carefully.
- Clamp off coolant hoses -2- with hose clips, up to 25 mm - 3094- -3-.
- Release hose clips -1- with spring-type clip pliers - VAS 5024A- or hose clip pliers - VAS 6340- .
- Place drip tray for workshop hoist - VAS 6208- underneath.
- Pull coolant hoses -2- off circulation pump - V55- -4-.
- Disconnect electrical connector -5-.
- Pull circulation pump - V55- -4- off bracket.



Installing

Install in reverse order of removal, observing the following:

- Fill up with coolant ⇒ Rep. gr. 19 ; Cooling system/coolant; Draining and filling coolant .
- Bleed cooling system ⇒ Rep. gr. 19 ; Cooling system, coolant; Draining and filling with coolant .



2.7 Removing and installing exhaust system

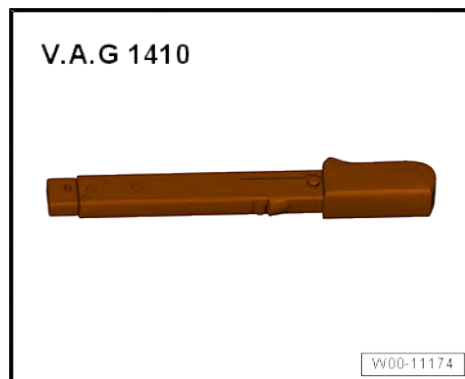
⇒ [“2.7.1 Removing and installing exhaust system, auxiliary coolant heater”, page 32](#)

⇒ [“2.7.2 Removing and installing exhaust system, auxiliary air heater”, page 32](#)

2.7.1 Removing and installing exhaust system, auxiliary coolant heater

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1410-



Removing

- Remove front left underbody cladding ⇒ General body repairs, exterior; Rep. gr. 66 ; Underbody cladding; Removing and installing underbody cladding .
- Remove front centre heat shield ⇒ General body repairs, exterior; Rep. gr. 66 ; Mouldings/trims/extensions; Removing and installing floor heat shield .

CAUTION

Danger of severe burns from hot exhaust system.

Burns to hands and other body parts possible.

- Allow exhaust system to cool.

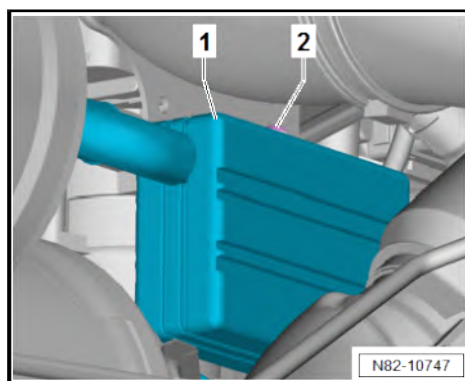
- Unscrew bolt -2-.
- Pull exhaust system -1- off auxiliary heater.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“2.2 Assembly overview - auxiliary heater”, page 14](#)



2.7.2 Removing and installing exhaust system, auxiliary air heater

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1410-

V.A.G 1410



WV00-11174

Removing

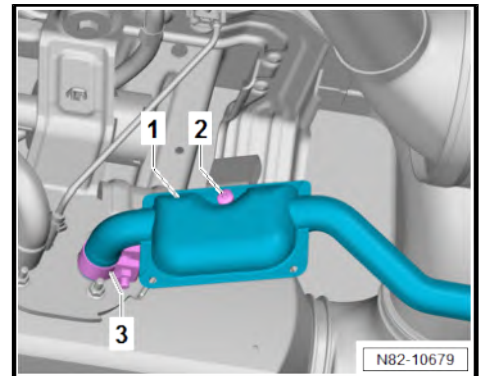
- If fitted, remove front right underbody cladding ⇒ General body repairs, exterior; Rep. gr. 66 ; Underbody cladding; Removing and installing underbody cladding .



CAUTION

Danger of severe burns from hot exhaust system.
Burns to hands and other body parts possible.
– Allow exhaust system to cool.

- Unscrew bolt -2-.
- Loosen screw-type clip -3-.
- Detach exhaust system -1-.



N82-10679

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“2.2 Assembly overview - auxiliary heater”, page 14](#)

2.8 Removing and installing auxiliary/supplementary heater bracket

⇒ [“2.8.1 Removing and installing auxiliary heater bracket, auxiliary coolant heater”, page 33](#)

⇒ [“2.8.2 Removing and installing auxiliary heater bracket, auxiliary air heater, heater unit”, page 35](#)

⇒ [“2.8.3 Removing and installing auxiliary heater bracket, auxiliary air heater, body”, page 35](#)

2.8.1 Removing and installing auxiliary heater bracket, auxiliary coolant heater

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1410-

V.A.G 1410



W00-11174

Removing

- Auxiliary heater removed ⇒ [page 17](#)
- Press together spreader clip -2-.
- Detach air intake silencer -1-.

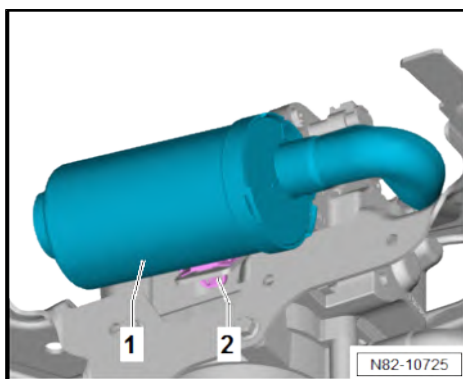
- Unscrew bolt -1-.
- Detach exhaust system -2-.



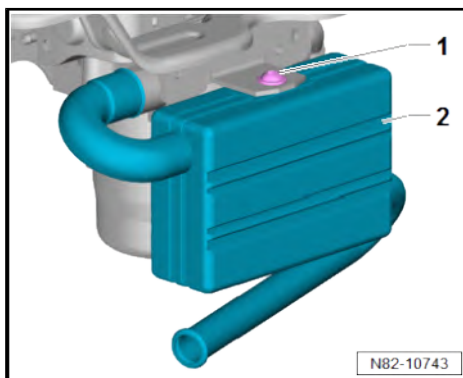
Note

Mark coolant hoses prior to removal.

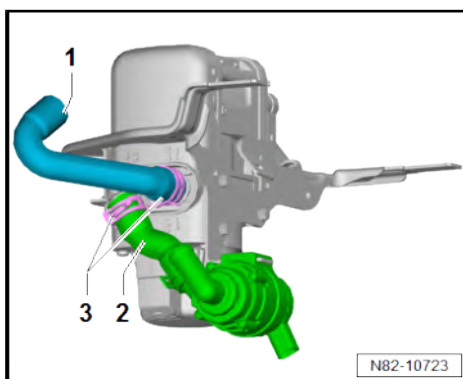
- Loosen spring-type clips -3-.
- Pull off coolant hose -1-.
- Pull off coolant hose -2- together with circulation pump - V55- .



N82-10725



N82-10743



N82-10723



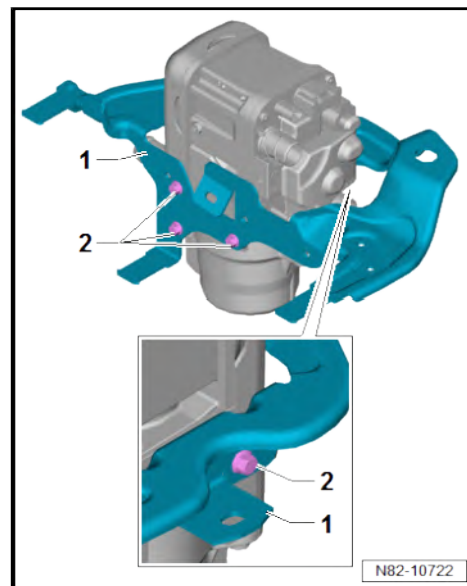
- Unscrew bolts -2-.
- Detach bracket -1- from auxiliary heater.

Installing

Install in reverse order of removal, observing the following:

Specified torques

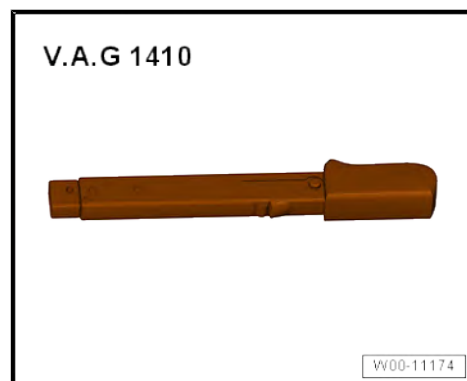
- ◆ ⇒ [“2.2 Assembly overview - auxiliary heater”, page 14](#)



2.8.2 Removing and installing auxiliary heater bracket, auxiliary air heater, heater unit

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1410-



Removing

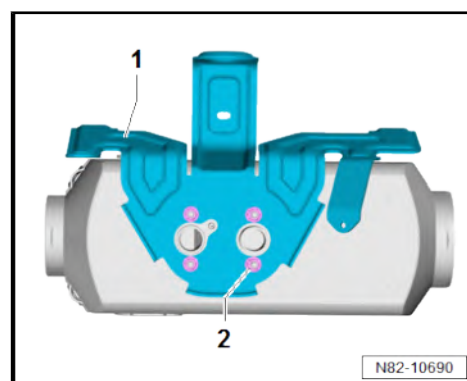
- Remove auxiliary air heater ⇒ [page 19](#) .
- Remove air intake silencer ⇒ [page 30](#) .
- Remove exhaust system ⇒ [page 32](#) .
- Unscrew nuts -2-.
- Remove bracket -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“2.2 Assembly overview - auxiliary heater”, page 14](#)

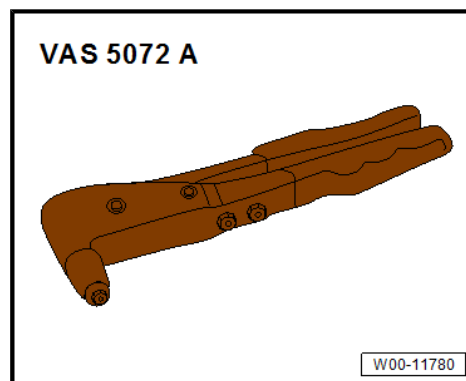


2.8.3 Removing and installing auxiliary heater bracket, auxiliary air heater, body

Special tools and workshop equipment required



- ◆ Pop rivet pliers - VAS 5072 A-



- ◆ Drill

Removing

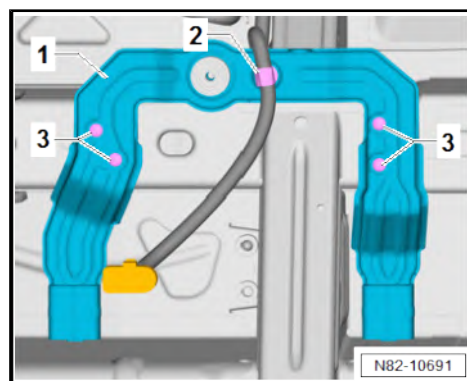
- Remove auxiliary air heater ➔ [page 19](#) .
- Unclip bracket -2- from damper element holder -1-.

⚠ CAUTION

Risk of injury from swarf being flung into air.

Risk of irritation and injury to skin and eyes.

- Wear protective goggles.
- Wear protective gloves.



Note

The following work steps must be performed by a second mechanic.

- Have a second mechanic secure damper element holder -1-.
- Drill out pop rivets -3- with a drill .
- Remove damper element holder -1-.

Installing

Install in reverse order of removal, observing the following:

- Install pop rivets using pop rivet pliers - VAS 5072 A- .



2.9 Removing and installing cover

⇒ ["2.9.1 Removing and installing cover, auxiliary heater control unit J364", page 37](#)

⇒ ["2.9.2 Removing and installing cover, burner unit", page 37](#)

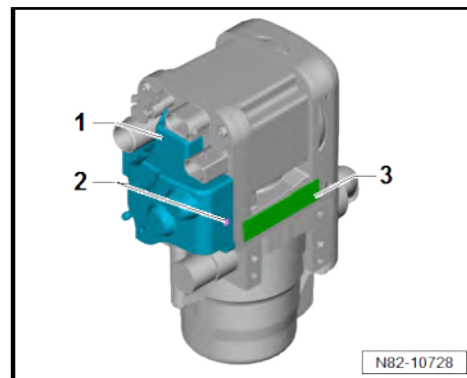
2.9.1 Removing and installing cover, auxiliary heater control unit - J364-

Removing

- Remove auxiliary heater bracket ⇒ [page 33](#) .
- Unclip cover -1- in area -2-.
- Remove covers -1- and -3-.

Installing

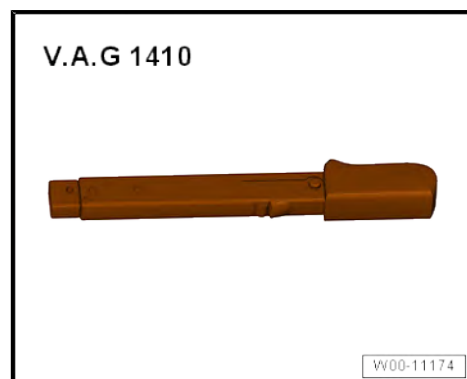
Install in reverse order of removal.



2.9.2 Removing and installing cover, burner unit

Special tools and workshop equipment required

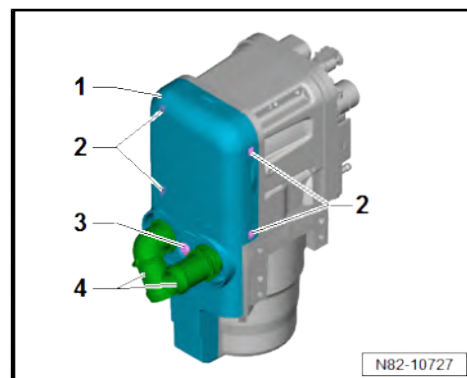
- ◆ Torque wrench - V.A.G 1410-



Removing

- Remove auxiliary heater bracket ⇒ [page 33](#) .
- Unscrew bolt -3-.
- Detach coolant connections together with bracket -4-.
- Unclip cover -1- in area -2-.
- Take off cover -1-.

Installing

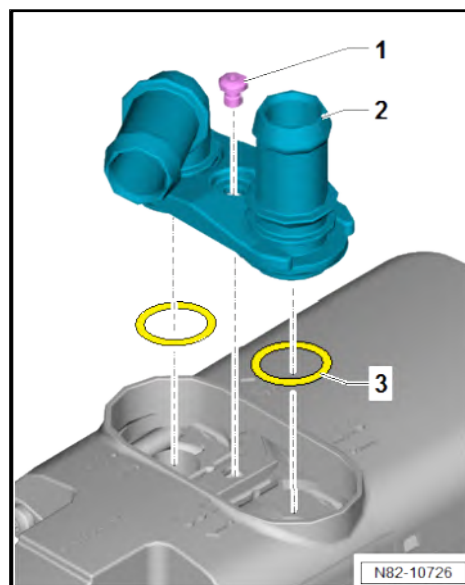




Install in reverse order of removal, observing the following:

- Check sealing surfaces for soiling and damage.
- Moisten oil seals with coolant before installing coolant connections.
- Renew oil seals -3-.
- Insert coolant connections together with bracket -2-.
- Tighten bolt -1-.

Specified torques

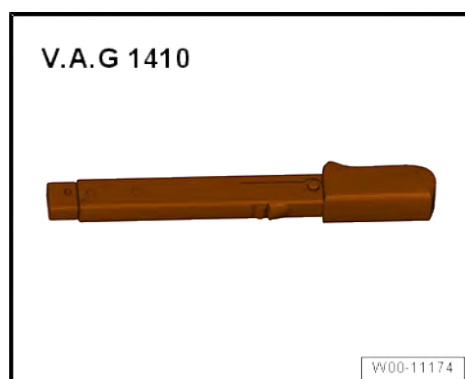


Component	Specified torque
Bolt	4 Nm

2.10 Dismantling and assembling heater unit

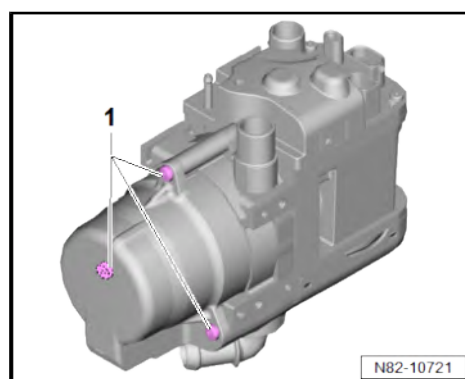
Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1410-



Dismantling

- Remove cover ⇒ [page 37](#) .
- Unscrew bolts -1-.
- Disconnect electrical connectors from temperature sensor - G18- / overheating sensor - G189- .
- Pull heat exchanger off blower housing.





- Detach gasket -1- from blower housing.
- Disconnect electrical connector -3-.
- Remove battery unit.

Installing

Assemble in reverse order of dismantling observing the following:

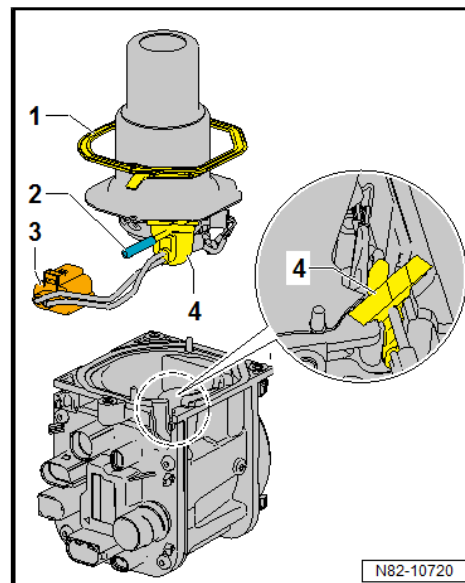


Note

Make sure that the gasket -1-, the grommet -4- and the fuel connection -2- are positioned correctly.

- Clean sealing surfaces.
- Renew seal -1-.

Specified torques



Component	Specified torque
Bolts	7 Nm

2.11 Removing and installing glow plug with flame monitor - Q8-

Removing

- Dismantling and assembling heater unit ⇒ [page 38](#) .
- Pull out retaining spring -1- in direction of -arrow-.
- Remove retaining plate -2- in direction of -arrow-.
- Pull out glow plug with flame monitor - Q8- -3-.

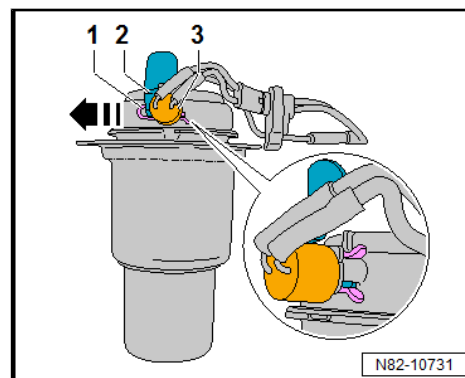
Installing

Install in reverse order of removal, observing the following:



Note

- ◆ *Observe installation position of retaining spring -1- and retaining plate -2-.*
- ◆ *Curvature of retaining spring -1- faces up.*





2.12 Removing and installing temperature sensor - G18- / overheating sensor - G189-

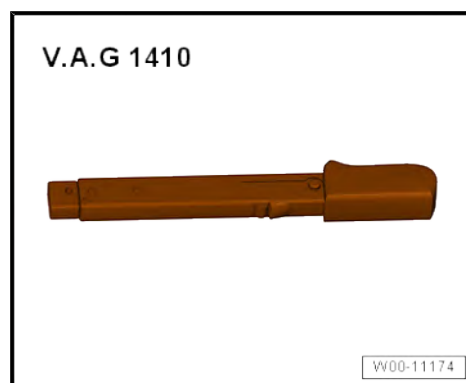
⇒ [“2.12.1 Removing and installing temperature sensor G18 / overheating sensor G189 , auxiliary coolant heater”, page 40](#)

⇒ [“2.12.2 Removing and installing temperature sensor G18 / overheating sensor G189 , auxiliary air heater”, page 41](#)

2.12.1 Removing and installing temperature sensor - G18- / overheating sensor - G189- , auxiliary coolant heater

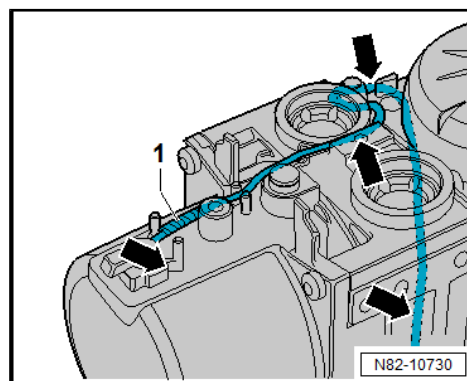
Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1410-



Removing

- Remove cover ⇒ [page 37](#) .
- Unclip electrical wire -1- from retainers -arrows-.





- Unscrew bolt -2-.
- Remove retaining spring -1-.
- Pull out temperature sensor - G18- -4- and overheating sensor - G189- -3- using long-nose pliers.

Installing

Install in reverse order of removal, observing the following:

- Coat seals from temperature sensor - G18- and overheating sensor - G189- with coolant for installation.

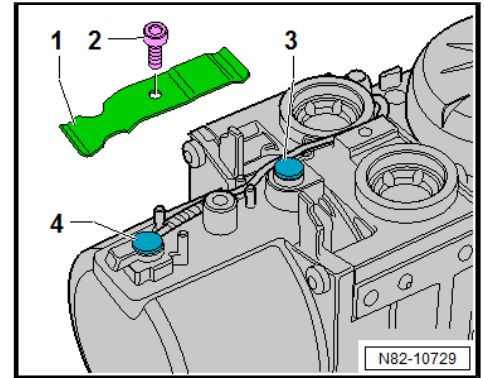


Note

- ◆ *When assembling, ensure proper position of retaining spring -1-.*
- ◆ *Sensors cannot be renewed individually.*

Specified torques

Component	Specified torque
Bolt -2-	4 Nm



2.12.2 Removing and installing temperature sensor - G18- / overheating sensor - G189- , auxiliary air heater

The temperature sensor - G18- is an integral part of the auxiliary air heater control unit - J604- -1- and cannot be renewed individually ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

2.13 Removing and installing auxiliary heater control unit - J364-

The auxiliary heater control unit - J364- is an integral part of the combustion air blower - V6- and cannot be renewed individually ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

2.14 Removing and installing combustion air blower - V6-

Removing

Dismantling and assembling heater unit ⇒ [page 38](#) .

Installing

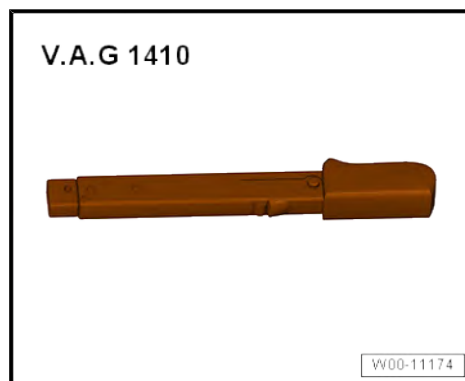
Install in reverse order of removal.

2.15 Removing and installing distribution box, auxiliary air heater

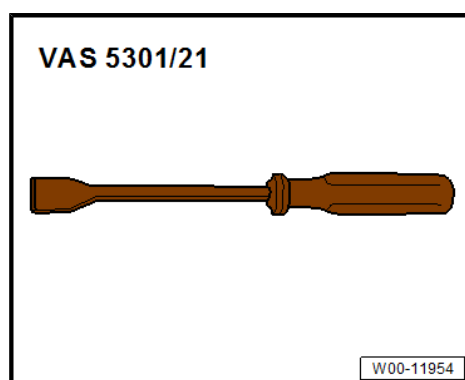
Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1410-



- ◆ Scraper - VAS 5301/21-



Removing

- Remove floor covering on right in passenger compartment ⇒ General body repairs, interior; Rep. gr. 70 ; Trims, interior; Removing and installing floor covering .

Vehicles with two heater units

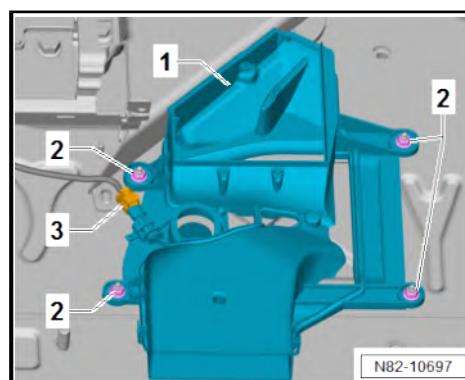
- Remove air duct on second heater unit ⇒ Heater and fresh air ventilation system; Rep. gr. 87 ; Air duct system; Removing and installing air ducts; Removing and installing air duct on second heater unit .

Continued for all vehicles

- Remove vent duct ⇒ [page 43](#) .
- Unscrew nuts -2-.
- Disconnect connector -3-.
- Remove distribution box for auxiliary air heater -1-.

Installing

Install in reverse order of removal, observing the following:

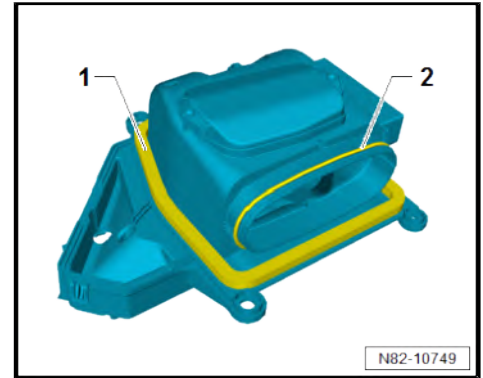




Note

The distribution box seal -1- cannot be removed intact.

- Remove distribution box seal -1- from distribution box using scraper - VAS 5301/21- .
- Clean bonding surface of distribution box seal -1- on distribution box.
- Renew distribution box seal -1- ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.
- Inspect seal -2- for damage, and renew it if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.



Specified torques

Component	Specified torque
Nuts -2-	8 Nm

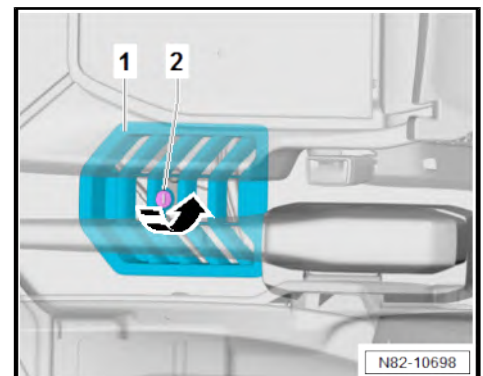
2.16 Removing and installing bottom vents, auxiliary air heater

Removing

- Turn rotary fastener -2- by 90° in direction of -arrow-.
- Swing bottom vent for footwell -1- in front area upwards, and detach it from distribution box for auxiliary air heater.

Installing

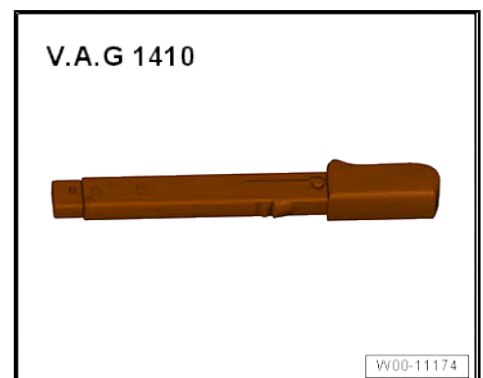
Install in reverse order of removal.



2.17 Removing and installing vent duct for auxiliary air heater

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1410-





Removing

- Remove front right underbody cladding ⇒ General body repairs, exterior; Rep. gr. 66 ; Underbody cladding; Removing and installing underbody cladding .

CAUTION

Danger of severe burns from hot exhaust system.
Burns to hands and other body parts possible.

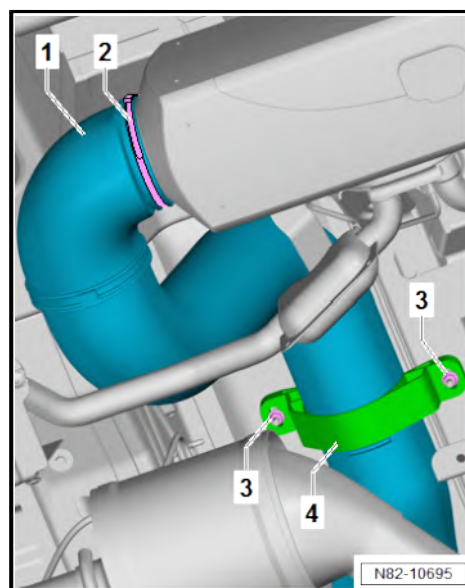
- Allow exhaust system to cool.

Vehicles with PR number 2D0

- Remove stone deflector ⇒ [page 46](#) .

Continued for all vehicles

- Remove cable tie -2-.
- Unscrew nuts -3-. While doing this, detach bracket -4-.
- Pull vent duct -1- off heater unit.
- Pull vent duct -1- off air distribution box, and remove it downwards.



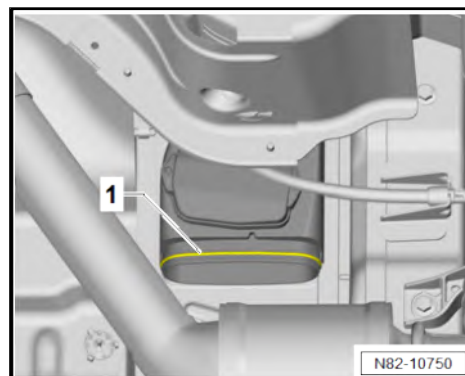
Installing

Install in reverse order of removal, observing the following:

- Inspect seal -1- for damage, and renew it if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

Specified torques

- ♦ ⇒ [“2.2 Assembly overview - auxiliary heater”, page 14](#)

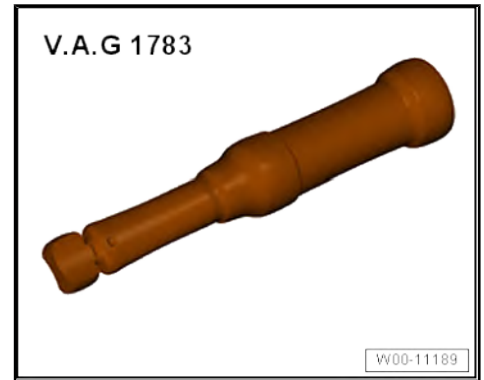


2.18 Removing and installing intake port for auxiliary air heater

Special tools and workshop equipment required

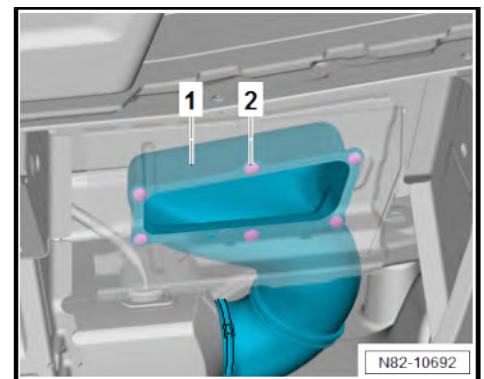


- ◆ Torque wrench - V.A.G 1783-



Removing

- Remove right step panel ⇒ General body repairs, interior; Rep. gr. 70 ; Trims, interior; Removing and installing step panel .
- Unscrew bolts -2-.
- Detach intake port -1- from step panel.
- Remove front right underbody cladding ⇒ General body repairs, exterior; Rep. gr. 66 ; Underbody cladding; Removing and installing underbody cladding .

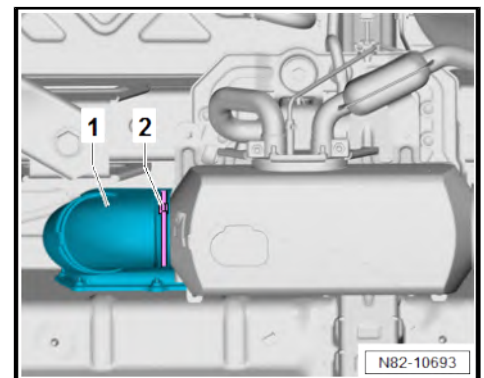


Vehicles with PR number 2D0

- Remove stone deflector ⇒ [page 46](#) .

Continued for all vehicles

- Remove cable tie -2-.
- Pull intake port -1- off heater unit, and remove it downwards.

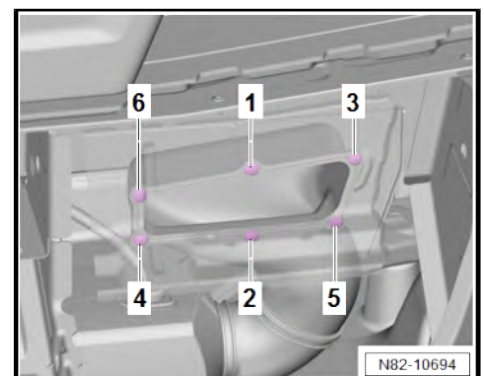


Installing

Install in reverse order of removal, observing the following:

- Tighten bolts -1 to 6- in the sequence shown.

Specified torques



Component	Specified torque
Bolts -1 to 6-	3.5 Nm



2.19 Removing and installing stone deflector for auxiliary air heater

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1410-



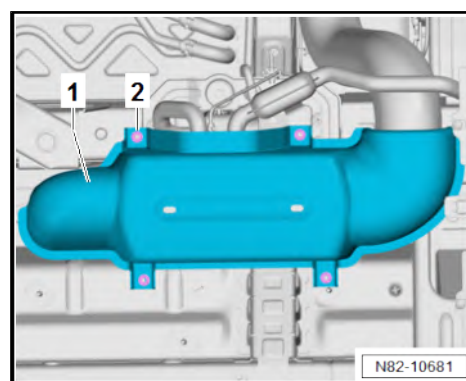
Removing

- Unscrew nuts -2-.
- Remove stone deflector -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques



Component	Specified torque
Nuts -2-	8 Nm

2.20 Removing and installing bracket for vent duct of auxiliary air heater

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1410-





Removing

- Remove front right underbody cladding ⇒ General body repairs, exterior; Rep. gr. 66 ; Underbody cladding; Removing and installing underbody cladding .

Vehicles with PR number 2D0

- Remove stone deflector ⇒ [page 46](#) .

Continued for all vehicles

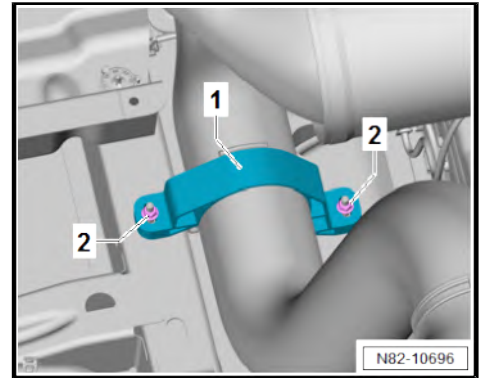
- Unscrew nuts -2-. While doing this, detach bracket -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

- ◆ ⇒ [“2.2 Assembly overview - auxiliary heater”, page 14](#)



3 Coolant circuit with auxiliary/supplementary heater

⇒ ["3.1 Connection diagram - coolant hoses", page 48](#)

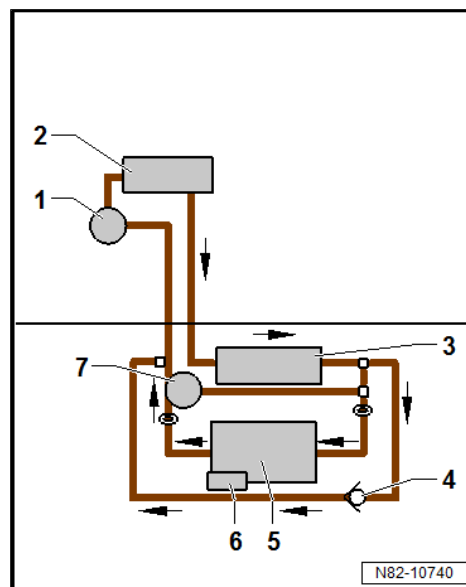
⇒ ["3.2 Bleeding coolant circuit of auxiliary/supplementary heater", page 49](#)

⇒ ["3.3 Removing and installing heater coolant shut-off valve", page 50](#)

3.1 Connection diagram - coolant hoses

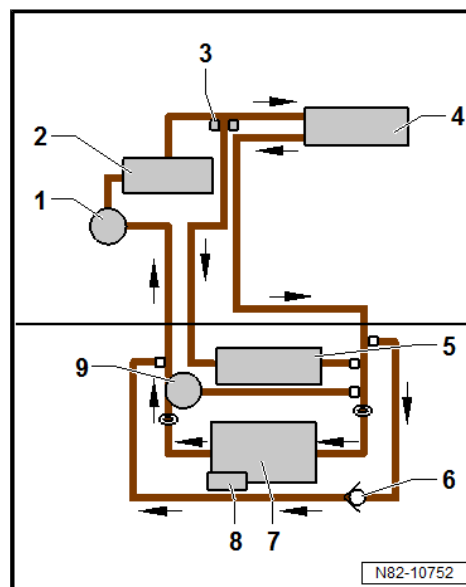
Auxiliary heater with one heat exchanger

- 1 - Circulation pump - V55-
- 2 - Auxiliary/supplementary heater
- 3 - Heat exchanger
- 4 - Non-return valve
- 5 - Engine
- 6 - Auxiliary pump for heating - V488-
- 7 - 3/2 way valve



Auxiliary heater with two heat exchangers

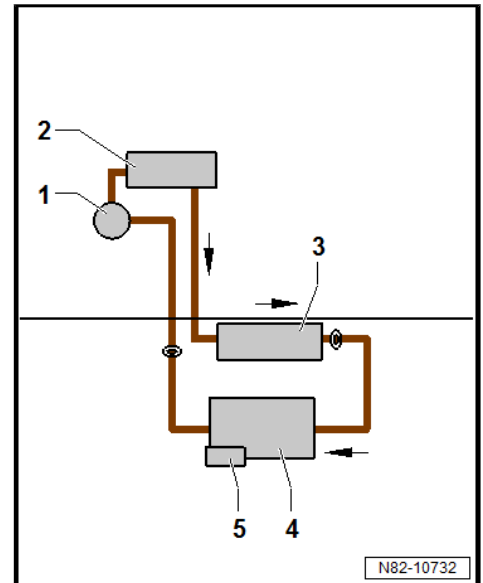
- 1 - Circulation pump - V55-
- 2 - Auxiliary/supplementary heater
- 3 - Thermostat
- 4 - Second heat exchanger
- 5 - Heat exchanger
- 6 - Non-return valve
- 7 - Engine
- 8 - Auxiliary pump for heating - V488-
- 9 - 3/2 way valve





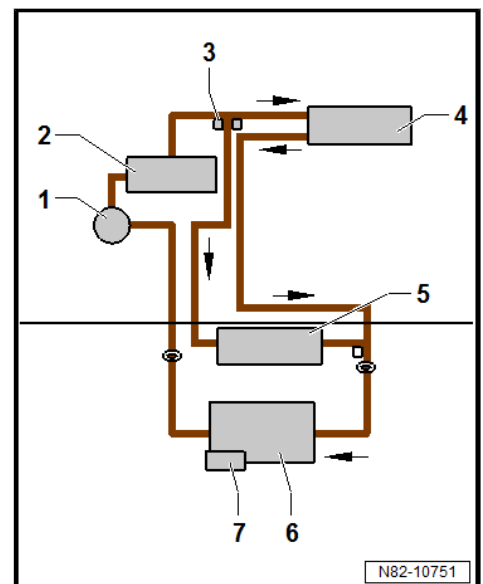
Supplementary heater

- 1 - Circulation pump - V55-
- 2 - Supplementary heater
- 3 - Heat exchanger
- 4 - Engine
- 5 - Auxiliary pump for heating - V488-



Supplementary heater with two heat exchangers

- 1 - Circulation pump - V55-
- 2 - Supplementary heater
- 3 - Thermostat
- 4 - Second heat exchanger
- 5 - Heat exchanger
- 6 - Engine
- 7 - Auxiliary pump for heating - V488-



3.2 Bleeding coolant circuit of auxiliary/supplementary heater

Coolant circuit must be bled after removal and installation of auxiliary heater/supplementary heater and circulation pump - V55- → Rep. gr. 19 ; Cooling system, coolant; Draining and filling coolant .

If a large quantity of coolant has escaped due to another problem, e.g. a leaking hose, or after coolant pipes have been renewed, the coolant circuit must be bled → Rep. gr. 19 ; Cooling system, coolant; Draining and filling coolant .

After bleeding and filling the coolant circuit, additional bleeding of the auxiliary/supplementary heater and the circulation pump - V55- must be performed as described below.

Bleeding coolant circuit of auxiliary/supplementary heater

- Activate circulation pump - V55- using ⇒ Vehicle diagnostic tester. This will bleed coolant circuit from heat exchanger in heater and air conditioning unit to circulation pump - V55- additionally.
- Start engine, and choose the highest available setting for temperature ("HI") on operating and display unit of heater and air



conditioning system. On vehicles equipped with Climatronic, turn rotary knobs to "HI" position (for driver and front passenger sides).

- Run auxiliary heater at full load for at least 2 minutes (circulation pump - V55- is activated).
- Run engine at approximately 2000 to 2500 rpm for at least 2 minutes.
- Switch off engine and auxiliary heater.

CAUTION

On a warm engine, the cooling system is under high pressure. Danger of scalding by steam and hot coolant.

Skin and other parts of the body may be scalded.

- Wear protective gloves.
 - Wear protective goggles.
 - Reduce excess pressure by covering cap of coolant expansion tank with cloths and opening it carefully.
-
- Check coolant level in coolant expansion tank, and top up coolant if necessary ⇒ Rep. gr. 19 ; Cooling system/coolant; Draining and filling coolant .

3.3 Removing and installing heater coolant shut-off valve

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1410-

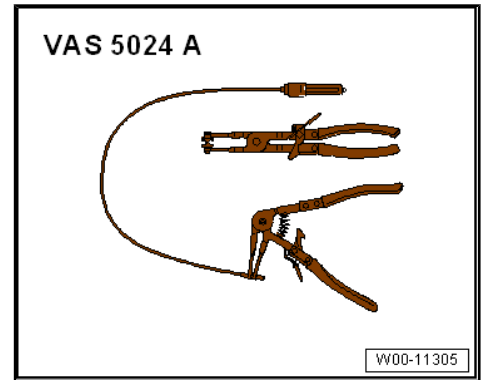


- ◆ Hose clamps to 25 mm - 3094-





- ◆ Spring-type clip pliers - VAS 5024A- or hose clip pliers - VAS 6340-



- ◆ Drip tray for workshop hoist - VAS 6208-



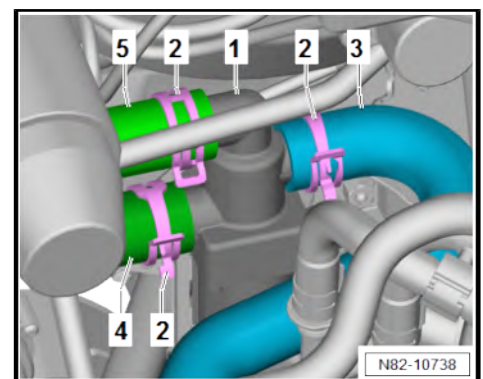
Removing

- Clamp off coolant hoses -3-, -4- and -5- using hose clamps for hoses up to 25 mm - 3094- .
- Release spring-type clips -2- with spring-type clip pliers - VAS 5024A- or hose clip pliers - VAS 6340- .
- Place drip tray for workshop hoist - VAS 6208- underneath.

⚠ CAUTION

**Danger of scalding by steam and hot coolant.
Skin and other parts of the body may be scalded.**

- Release pressure by pressing pressure relief valve on cooling system tester until the gauge shows a pressure of 0.



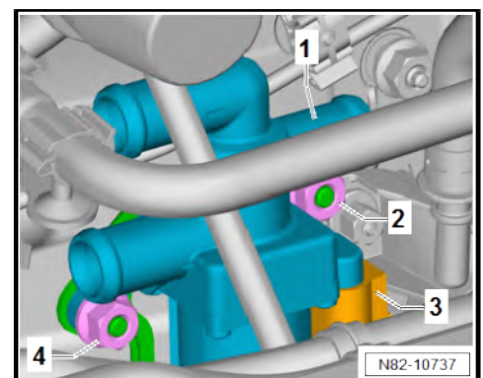
- Pull hoses -3-, -4- and -5- off shut-off valve -1-.
- Disconnect electrical connector -3-.
- Remove nuts -2- and -4-.
- Remove shut-off valve -1-.

Installing

Install in reverse order of removal, observing the following:

- Fill up with coolant ⇒ Rep. gr. 19 ; Cooling system/coolant; Draining and filling coolant .
- Bleed cooling system ⇒ Rep. gr. 19 ; Cooling system, coolant; Draining and filling with coolant .

Specified torques





Component	Specified torque
Nuts	8 Nm



4 Fuel supply system

⇒ [“4.1 Overview of fitting locations - fuel supply system”, page 53](#)

⇒ [“4.2 Checking fuel delivery rate”, page 54](#)

⇒ [“4.3 Removing and installing metering pump V54 ”, page 55](#)

⇒ [“4.4 Removing and installing fuel pump”, page 56](#)

⇒ [“4.5 Removing and installing fuel line leading to metering pump V54 ”, page 57](#)

⇒ [“4.6 Fuel line to fuel pump - heater unit for auxiliary air heater, removing and installing”, page 59](#)

⇒ [“4.7 Removing and installing bracket for metering pump V54 / fuel pump”, page 60](#)

⇒ [“4.8 Removing and installing intermediate bracket for metering pump V54 /fuel pump”, page 61](#)

4.1 Overview of fitting locations - fuel supply system

1 - Metering pump - V54-

- ☐ For auxiliary coolant heater
- ☐ Removing and installing
⇒ [page 55](#)

2 - Fuel pump

- ☐ For auxiliary air heater
- ☐ Removing and installing
⇒ [page 56](#)

3 - Fuel line

- ☐ Fuel pump - heater unit for auxiliary air heater
- ☐ Removing and installing
⇒ [page 59](#)

4 - Fuel line

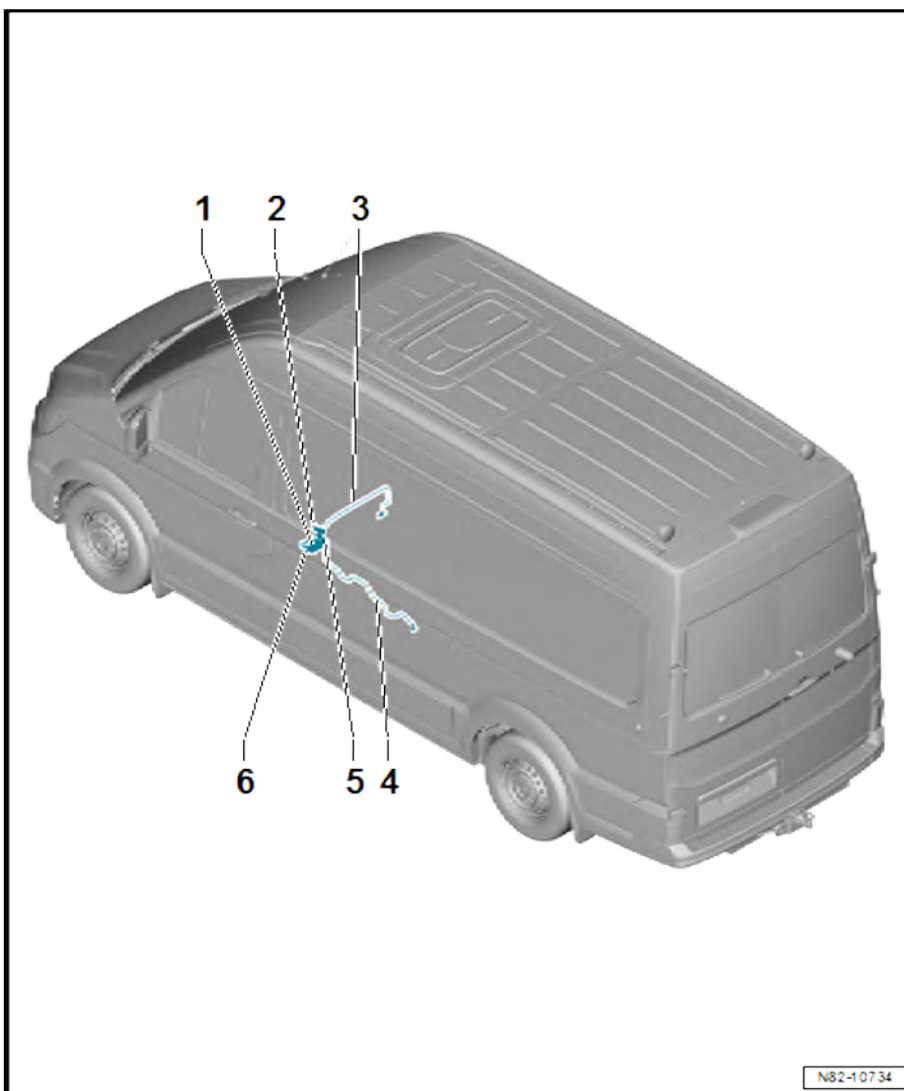
- ☐ To metering pump - V54-
- ☐ Removing and installing
⇒ [page 57](#)

5 - Bracket

- ☐ For metering pump - V54-
- ☐ Removing and installing
⇒ [page 60](#)

6 - Intermediate bracket

- ☐ For metering pump - V54-
- ☐ Removing and installing
⇒ [page 61](#)





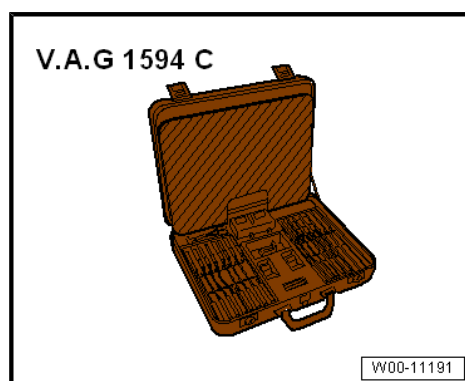
4.2 Checking fuel delivery rate

Special tools and workshop equipment required

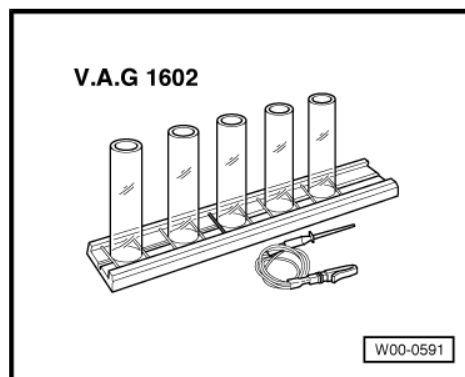
- ◆ Hand multimeter - V.A.G 1526 E-



- ◆ Auxiliary measuring set - V.A.G 1594C-



- ◆ Measuring beaker - V.A.G 1602/1- from injection rate tester - V.A.G 1602-



- ◆ Vehicle diagnostic tester

Test conditions

- Resistance of metering pump: 5.4 Ω
- Battery voltage: at least 12.5 V
- No fault stored in event memory
- Fuel lines are not damaged or leaking
- Ambient temperature approx. 20°C
- Fuel tank sufficiently filled (fuel gauge on dash panel insert not in red area).



Test sequence

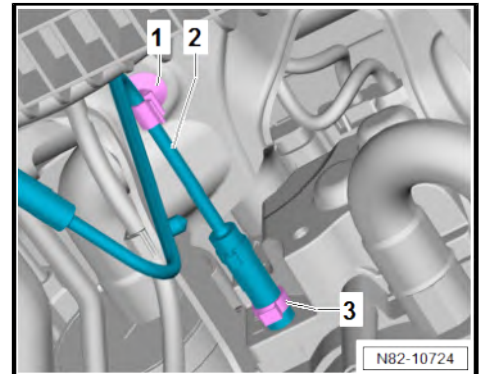
CAUTION

The fuel system is pressurised.
Danger of injury caused by fuel spray.

- Wear protective goggles.
- Wear protective gloves.
- To release pressure, wrap a clean cloth around the connection and carefully loosen the connection.

Vehicles with auxiliary coolant heater

- Unclip fuel line -2- from retainer -1-.
- Remove clamp -3-.
- Disconnect fuel line -2-.

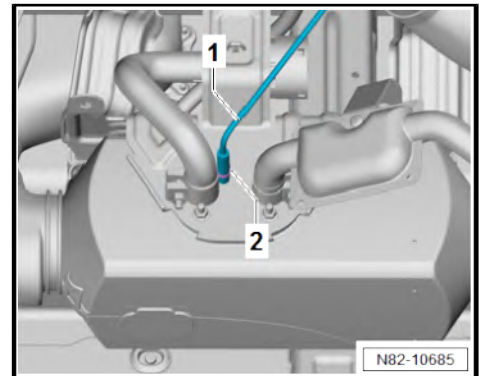


Vehicles with auxiliary air heater

- Remove clamp -2-.
- Pull off fuel line -1-.

Continued for all vehicles

- Guide fuel line -1- into measuring beaker - V.A.G 1602/1- .
- Hold measuring beaker - V.A.G 1602/1- at same height as auxiliary heater.
- Using ⇒ Vehicle diagnostic tester, check fuel delivery rate of metering pump - V54- .
- Read quantity of diesel in measuring beaker - V.A.G 1602/1- .



Vehicles with auxiliary coolant heater

Min. quantity of fuel after 60 seconds, ml	Max. quantity of fuel after 60 seconds, ml
16.2	17.8

Vehicles with auxiliary air heater

Min. quantity of fuel after 90 seconds, ml	Max. quantity of fuel after 90 seconds, ml
18	22

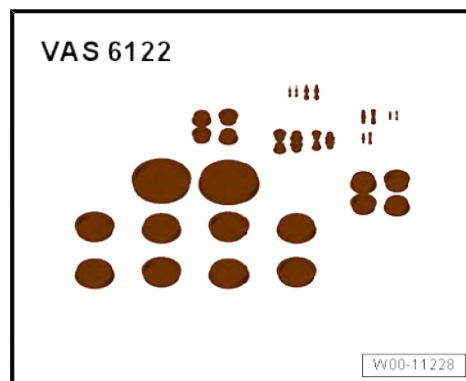
- If fuel delivery rates are not within specifications, renew metering pump - V54- ⇒ [page 55](#) .

4.3 Removing and installing metering pump - V54-

Special tools and workshop equipment required



- ◆ Set of plugs for engine - VAS 6122-



Removing

- Remove front left underbody cladding ⇒ General body repairs, exterior; Rep. gr. 66 ; Underbody cladding; Removing and installing underbody cladding .

CAUTION

Danger of severe burns from hot exhaust system.
Burns to hands and other body parts possible.

- Allow exhaust system to cool.

- Disconnect electrical connector -2-.
- Remove clamps -3-.

CAUTION

The fuel system is pressurised.
Danger of injury caused by fuel spray.

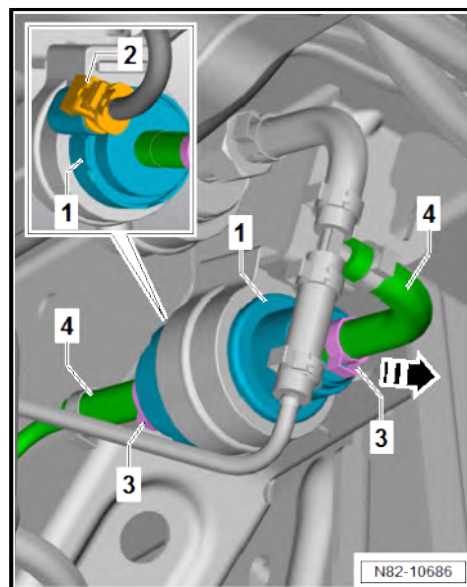
- Wear protective goggles.
- Wear protective gloves.
- To release pressure, wrap a clean cloth around the connection and carefully loosen the connection.

- Pull fuel hoses -4- off metering pump - V54- .
- Seal fuel hoses -4- with clean plugs from engine bung set - VAS 6122- .
- Remove metering pump - V54- -1- in -direction of arrow -.

Installing

Install in reverse order of removal, observing the following:

- Push metering pump - V54- -1- into bracket as far as stop at connector mounting.

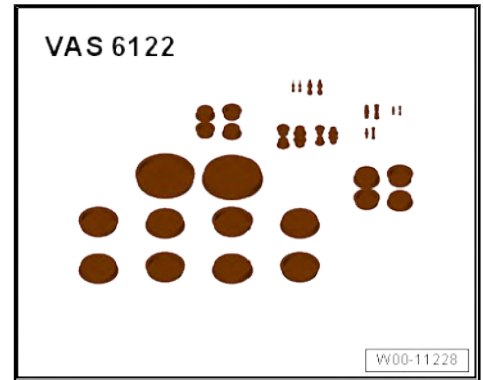


4.4 Removing and installing fuel pump

Special tools and workshop equipment required



- ◆ Set of plugs for engine - VAS 6122-



Removing

- Remove front left underbody cladding ⇒ General body repairs, exterior; Rep. gr. 66 ; Underbody cladding; Removing and installing underbody cladding .

CAUTION

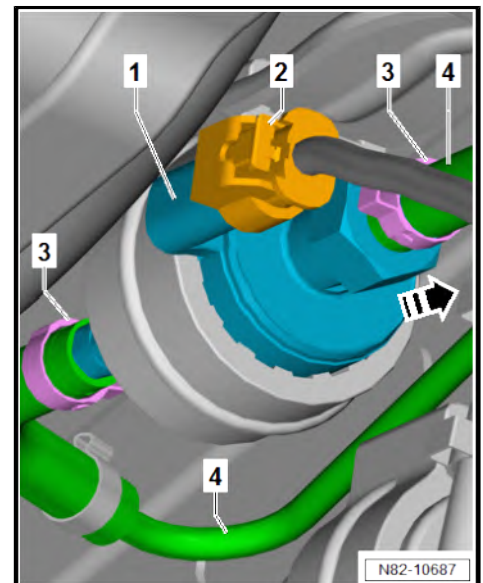
Danger of severe burns from hot exhaust system.
Burns to hands and other body parts possible.
– Allow exhaust system to cool.

- Disconnect electrical connector -2-.
- Remove clamps -3-.

CAUTION

The fuel system is pressurised.
Danger of injury caused by fuel spray.
– Wear protective goggles.
– Wear protective gloves.
– To release pressure, wrap a clean cloth around the connection and carefully loosen the connection.

- Pull fuel hoses -4- off fuel pump.
- Seal fuel hoses -4- with clean plugs from engine bung set - VAS 6122- .
- Remove fuel pump -1- in -direction of arrow-.



Installing

Install in reverse order of removal, observing the following:

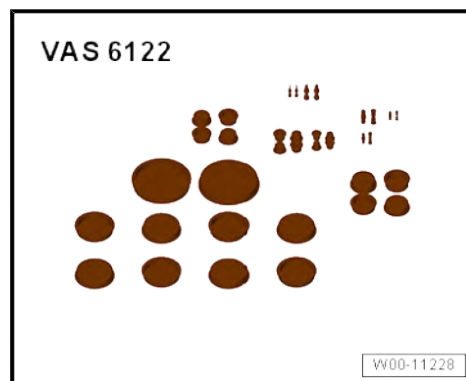
- Push fuel pump into bracket as far as stop at connector mounting.

4.5 Removing and installing fuel line leading to metering pump - V54-

Special tools and workshop equipment required



- ◆ Set of plugs for engine - VAS 6122-



- ◆ Drip tray for workshop hoist - VAS 6208-



Removing

- Remove front left underbody cladding ⇒ General body repairs, exterior; Rep. gr. 66 ; Underbody cladding; Removing and installing underbody cladding .

CAUTION

**Danger of severe burns from hot exhaust system.
Burns to hands and other body parts possible.**

- Allow exhaust system to cool.

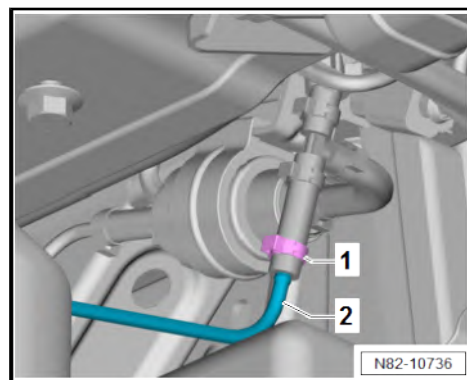
- Remove clamp -1-.

CAUTION

**The fuel system is pressurised.
Danger of injury caused by fuel spray.**

- Wear protective goggles.
- Wear protective gloves.
- To release pressure, wrap a clean cloth around the connection and carefully loosen the connection.

- Pull fuel line -2- off metering pump - V54- .
- Seal open connections with clean plugs from engine bung set - VAS 6122- .
- Lower fuel tank ⇒ Rep. gr. 20 ; Fuel tank; Removing and installing fuel tank .

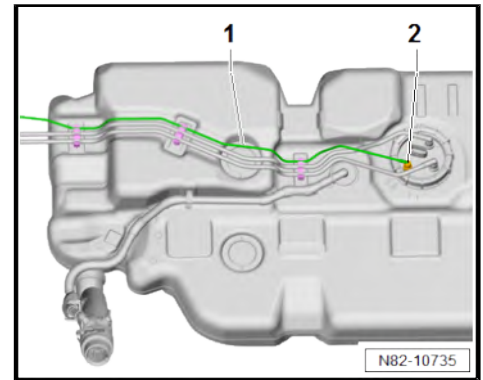




- Unclip fuel line -1- from retainers.
- Release and disconnect connection -2-.
- Remove fuel line -1-.

Installing

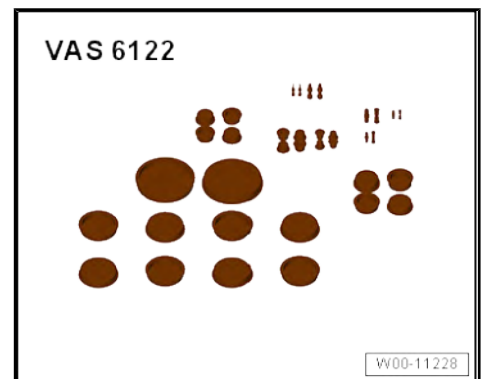
Install in reverse order of removal.



4.6 Fuel line to fuel pump - heater unit for auxiliary air heater, removing and installing

Special tools and workshop equipment required

- ◆ Set of plugs for engine - VAS 6122-



- ◆ Drip tray for workshop hoist - VAS 6208-



Removing

- Remove vent duct for auxiliary air heater ⇒ [page 43](#) .
- Remove front left underbody cladding ⇒ General body repairs, exterior; Rep. gr. 66 ; Underbody cladding; Removing and installing underbody cladding .

CAUTION

**Danger of severe burns from hot exhaust system.
Burns to hands and other body parts possible.**

- Allow exhaust system to cool.

- Place drip tray for workshop hoist - VAS 6208- underneath.



- Remove clamps -2-.

CAUTION

The fuel system is pressurised.

Danger of injury caused by fuel spray.

- Wear protective goggles.
 - Wear protective gloves.
 - To release pressure, wrap a clean cloth around the connection and carefully loosen the connection.
-
- Pull fuel line -1- off fuel pump and heater unit.
 - Pull fuel line -1- off heater unit.
 - Unclip fuel line -1- in area indicated by -arrows-, and guide it out towards bottom.
 - Seal open connections with clean plugs from engine bung set - VAS 6122- .

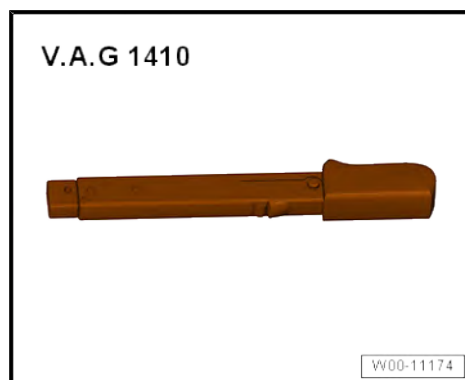
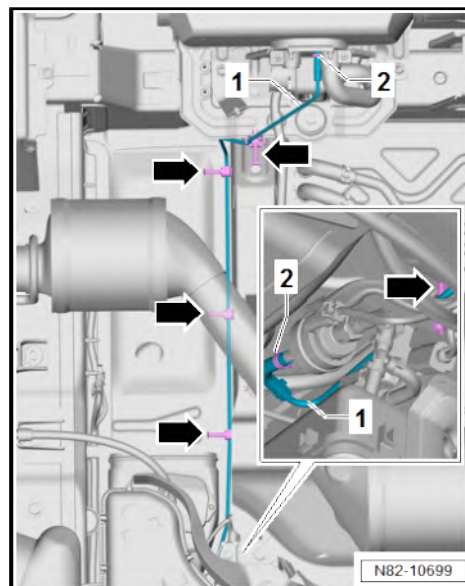
Installing

Install in reverse order of removal.

4.7 Removing and installing bracket for metering pump - V54- /fuel pump

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1410-





Removing

- Remove front left underbody cladding ⇒ General body repairs, exterior; Rep. gr. 66 ; Underbody cladding; Removing and installing underbody cladding .

CAUTION

Danger of severe burns from hot exhaust system.

Burns to hands and other body parts possible.

- Allow exhaust system to cool.

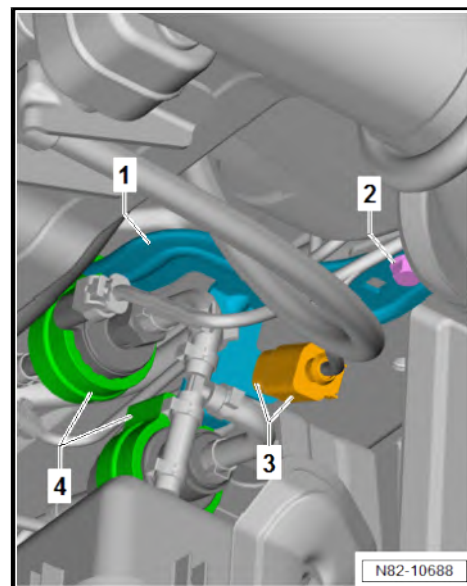
- Push bracket -4- together with metering pump - V54- and fuel pump off bracket -1-.
- Unclip electrical connector -3- from bracket -1-.
- Unscrew bolt -2-. While doing this, detach bracket -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques

Component	Specified torque
Bolt -2-	8 Nm



4.8 Removing and installing intermediate bracket for metering pump - V54- /fuel pump

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1410-

V.A.G 1410



W00-11174

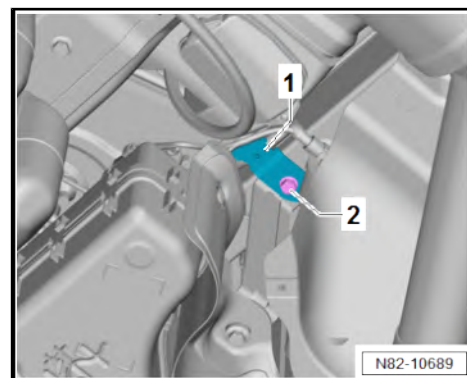
Removing

- Remove bracket for metering pump - V54- /fuel pump ⇒ [page 60](#) .
- Unscrew bolt -2-. While doing this, detach bracket -1-.

Installing

Install in reverse order of removal, observing the following:

Specified torques





Component	Specified torque
Bolt -2-	20 Nm



5 Regulation of auxiliary/supplementary heater

⇒ ["5.1 Functional description", page 63](#)

⇒ ["5.2 Supplementary heater mode", page 66](#)

⇒ ["5.3 Auxiliary heater mode", page 67](#)

⇒ ["5.4 Switching off auxiliary/supplementary heater", page 67](#)

5.1 Functional description

⇒ ["5.1.1 Functional description of auxiliary coolant heater", page 63](#)

⇒ ["5.1.2 Functional description of auxiliary air heater", page 64](#)

5.1.1 Functional description of auxiliary coolant heater

The auxiliary heater can be started by means of the remote control, the instant heating button or appropriate programming in the central monitor. The supplementary heater mode is started automatically by the vehicle electronics system once all the necessary preconditions have been met ⇒ [page 6](#) .

First, a functional check of the fresh air blower - V2- and the combined glow plug with flame monitor - Q8- is performed, and then the glow plug with flame monitor - Q8- goes through a preheating phase.

The subsequent starting sequence depends, amongst other things, on the coolant temperature; therefore the time required for the starting sequence varies. In addition, fuel is supplied at this point, the glow plug with flame monitor - Q8- is switched off in stages and the flame monitor is activated.

At end of start phase, heater unit burns at full load first and then stepless standard operation ensues.

The reference resistance of the glow plug with flame monitoring - Q8- is automatically re-determined during the starting or run-on phase, depending on the unit. This is done to compensate for ageing-related changes.

As a consequence of this calibrating procedure, processes that are affected can also require more time.

After expiry of maximum operating time of 120 minutes or after manual shutdown by means of remote control or instant heating button , auxiliary heater is switched off.

The supplementary heater is switched off if at least one of the defined switch-on criteria is no longer met ⇒ [page 6](#) .

After switch-off, operation of the metering pump - V54- is stopped immediately.

The combustion air blower - V6- and the glow plug with flame monitor - Q8- continue to operate for a limited time in order to burn the remaining fuel in the system and to cool the burner components.

The run-on period of the heater can vary depending on the software version and load level at the time of switch-off.



5.1.2 Functional description of auxiliary air heater



Note

- ♦ *Make sure that the auxiliary air heaters are operated only with diesel fuel. Operation with biodiesel can lead to malfunctions and damage.*
- ♦ *The heaters only have one electric motor. The motor pumps the heating and combustion air.*

1 - Electrical connection

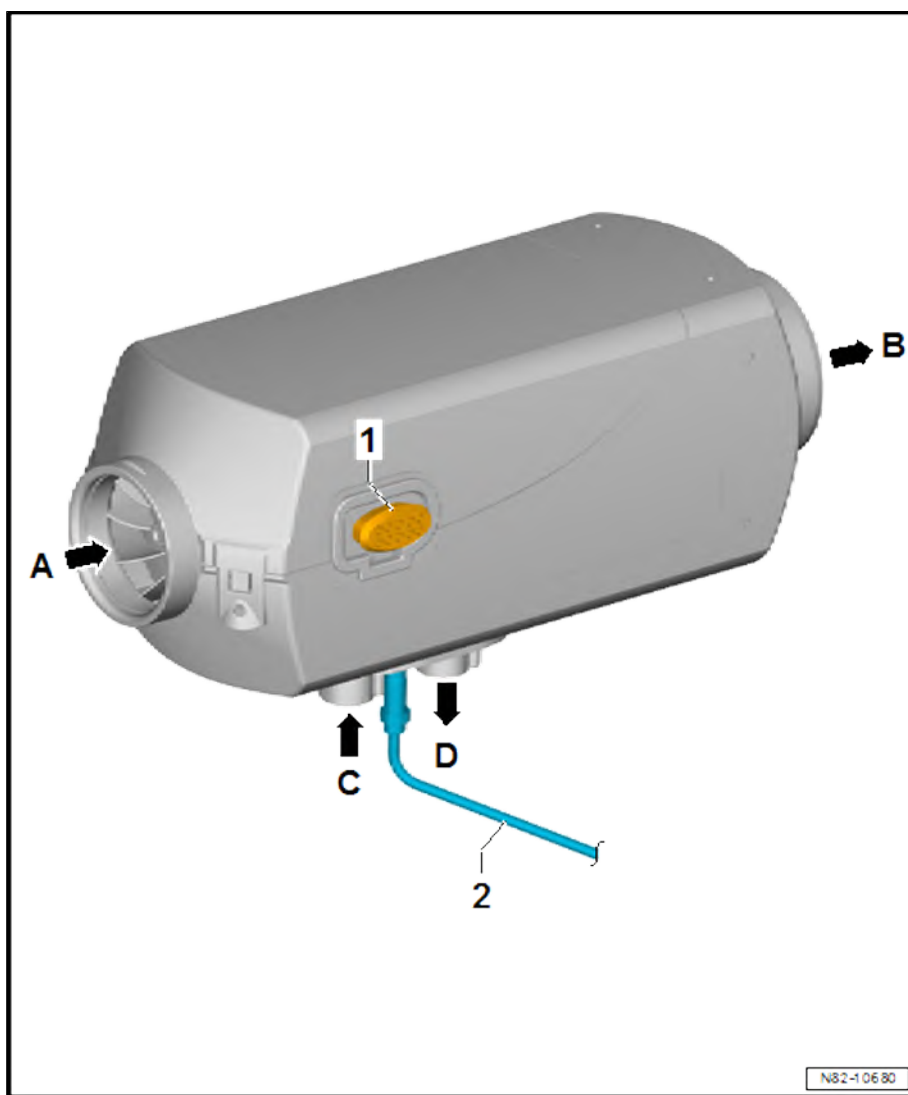
2 - Fuel supply system

Pfeil A - Cold air in

Pfeil B - Hot air out

Pfeil C - Combustion air in

Pfeil D - Exhaust out



Switching on

The warning lamp in the control element lights up when the unit is switched on. The glow plug is switched on and the combustion air blower runs at a slow speed (ventilation).



Note

- ◆ *If there is still too much residual heat in the heat exchanger from a previous heating operation, only the combustion air blower starts up at first (cold blowing).*
- ◆ *The »start« takes place once the existing residual heat has been dissipated.*

Airtronic start

Fuel delivery starts after about 60 seconds and the fuel/air mixture in the combustion chamber ignites. The glow plug is switched off 60 seconds after the combination sensor (flame monitor and overheating sensor) has detected the flame.

The heater is now in control mode and reaches the "Power" governor state after a short duration.

Temperature selection with control element

The desired interior temperature is preselected via the button. The value to be set can be chosen from a range of between +10° C and +30° C, depending on the version of the heater unit installed as well as the volume of the space to be heated and the prevailing ambient temperature.

The setting to be selected on the control knob in this case is based on experience.

Switching off

When the heater unit is switched off, the warning lamp goes out and the fuel delivery is switched off. The blower continues operating for about 4 minutes to provide cooling.

For cleaning, the glow plug is switched on for 40 seconds during the blower run-on.



Note

The heater unit is switched off without any run-on time if no fuel was pumped before the switch-off or if the heater unit is in the control level "OFF".

Control and safety devices of auxiliary air heater Airtronic D3

- ◆ The start is repeated if the heater unit does not ignite within 90 seconds after fuel delivery starts. A malfunction shut-off occurs if the heater does not ignite after another 90 seconds. That means that fuel delivery will be interrupted and blower run-on will continue for about 4 minutes. The control unit is locked out after the permitted number of unsuccessful start attempts has been exceeded.
- ◆ If the flame goes out by itself during heating operation, a new start is performed initially. If the heating unit does not ignite within 90 seconds after fuel delivery restarts or if it ignites but goes out within 15 minutes, a malfunction shut-off occurs. That means that fuel delivery will be interrupted and blower run-on will continue for about 4 minutes.



Note

- ◆ *The malfunction shut-down can be revoked by briefly switching off and back on again.*
- ◆ *Do not repeat the switch on/off procedure more than twice.*
- ◆ Overheating causes the combination sensor (flame monitor/overheating sensor) to be tripped, the fuel supply is interrupted and a malfunction shut-down is performed. Once the cause of overheating has been eliminated, the heater unit can be re-started by switching off and back on again. The control unit is locked out after the permitted number of unsuccessful start attempts has been exceeded.
- ◆ A malfunction shut-down will occur 20 seconds after the lower or upper voltage limit has been reached.
- ◆ The heater unit will not start if the glow plug or the blower motor is defective, or if there is a discontinuity in the electrical cable to the metering pump.
- ◆ If the combination sensor (flame monitor/overheating sensor) is defective or there is a discontinuity in the electrical cable, the heater unit starts and the malfunction shut-down occurs only during the start phase.
- ◆ The speed of the blower motor is continuously monitored. A malfunction shut-down will occur after 30 seconds if the blower motor does not start or if the speed fluctuates by more than 10%.
- ◆ When the heater unit is switched off, the glow plug is switched on for 40 seconds during blower run-on (post-glow) to burn off combustion residues.

Switching off in an emergency

If an emergency shut-off (emergency off) is required during operation, proceed as follows:

- Switch off heater unit via control element, or
- Pull out fuse, or
- Isolate heater unit from battery.

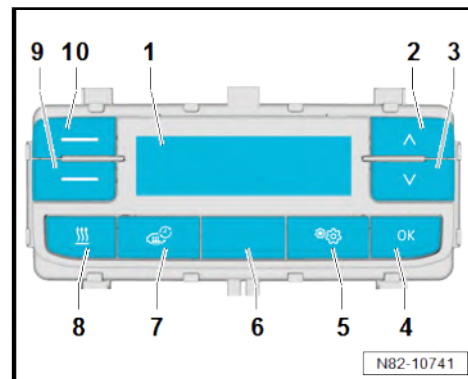
5.2 Supplementary heater mode

Once the coolant temperature has reached a certain level, the control unit switches to energy-saving mode. If the coolant temperature continues to rise, the control unit switches to regulated waiting period. If the coolant temperature does not drop during the regulated waiting period, however, the heater unit will perform the usual starting procedure and run in standard mode if coolant temperature drops below a certain level.



5.3 Auxiliary heater mode

- 1 - Display
- 2 - Increase ventilation
- 3 - Decrease ventilation
- 4 - Confirm
- 5 - Set-up menu
- 6 - Trim
- 7 - Auxiliary heater continuous operation
- 8 - Auxiliary heater on/off
- 9 - Increase temperature
- 10 - Decrease temperature



5.4 Switching off auxiliary/supplementary heater

If the instant heat button is pressed or the preset operating time has expired, combustion stops.

The run-on mode begins.

The circulation pump - V55- and the combustion air blower - V6- , however, continue to run to cool down the heater (run-on mode) and are switched off automatically.



Note

The run-on period of the circulation pump - V55- and the combustion air blower - V6- depends on the operating mode in which the auxiliary heater was switched off.

Run-on time is up to 4 minutes.

Depending on the control unit software version, slight deviations in the specified run-on values are possible.



6 Other controlling and regulating components

⇒ "6.1 Removing and installing ambient temperature sensor G17", page 68

⇒ "6.2 Removing and installing remote control receiver for auxiliary coolant heater R149", page 68

⇒ "6.3 Functional description of remote control for auxiliary/supplementary heater", page 69

⇒ "6.4 Removing and installing temperature sensor on roof for air inlet on auxiliary air heater G967", page 71

⇒ "6.5 Removing and installing remote control receiver for auxiliary heater R64", page 72

⇒ "6.6 Removing and installing air distribution flap control motor 2 V667", page 72

6.1 Removing and installing ambient temperature sensor - G17-



Note

The ambient temperature sensor - G17- is used for determining the outside temperature and as a switch-on condition of the auxiliary/supplementary heater.

- Removing and installing ambient temperature sensor - G17-
⇒ Heating, air conditioning; Rep. gr. 87 ; Other components for control and regulation; Removing and installing ambient temperature sender G17 .

6.2 Removing and installing remote control receiver for auxiliary coolant heater - R149-

Removing

Vehicles without storage compartment

- Remove moulded headliner ⇒ General body repairs, interior; Rep. gr. 70 ; Roof trims; Removing and installing moulded headliner .

Vehicles with storage compartment

- Remove storage compartment ⇒ General body repairs, interior; Rep. gr. 68 ; Compartments and covers; Removing and installing storage compartment in headliner .



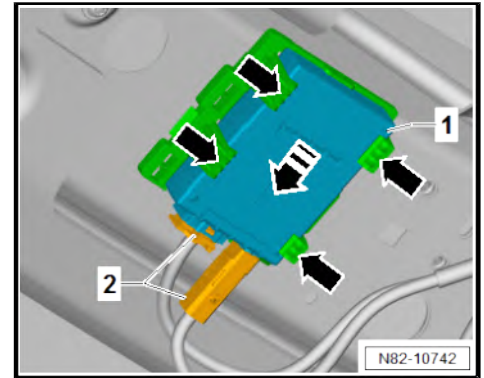
Continued for all vehicles

- Separate electrical connectors -2-.
- Detach remote control receiver for auxiliary coolant heater - R149- -1- in area indicted by -arrows-.
- Pull off remote control receiver for auxiliary coolant heater - R149- -1- in direction of -arrow-.

Installing

Install in reverse order of removal, observing the following:

- If necessary, remote control receiver for auxiliary coolant heater - R149- must be adapted to auxiliary heater/supplementary heater and remote control using ⇒ Vehicle diagnostic tester.



6.3 Functional description of remote control for auxiliary/supplementary heater



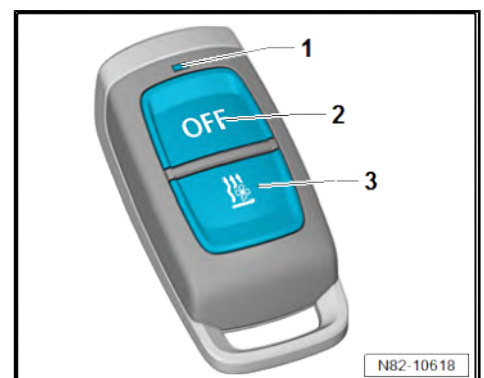
Note

- ◆ *The auxiliary heater can be switched on and off from a distance of several 100 metres (open space) using the remote control. This range is restricted in built-up areas or from within buildings.*
- ◆ *The remote control should be held vertically when the buttons are pressed. The signals are received best by the aerials in the vehicle in this position and the greatest range is achieved.*
- ◆ *Following disconnection of the vehicle batteries, the auxiliary/supplementary heater remote control does not have to be re-adapted.*
- ◆ *If the remote control does not work properly, check the remote control battery ⇒ [page 71](#) .*
- ◆ *If the remote control is renewed, it must be adapted using ⇒ Vehicle diagnostic tester.*
- ◆ *If the remote control is renewed, the remote control must be adapted to the remote control receiver for auxiliary coolant heater - R149- using ⇒ Vehicle diagnostic tester.*

Switching on auxiliary/supplementary heater using remote control

- Press button -3-. The indicator lamp -1- must light up green.

When the auxiliary heater is switched on, its operating time is determined by means of the setting on the multifunction indicator.



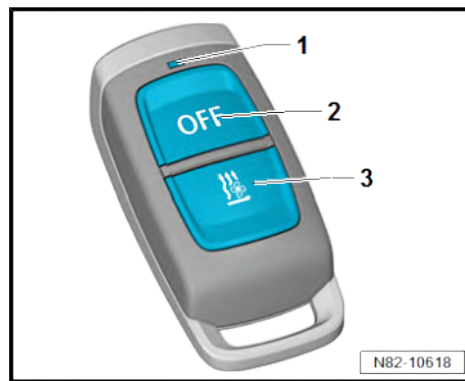


Switching off auxiliary/supplementary heater using remote control

- Press button -2-. The indicator lamp -1- must light up red.

The auxiliary and supplementary heater switches to run-on mode and then switches off.

Renewing remote control battery



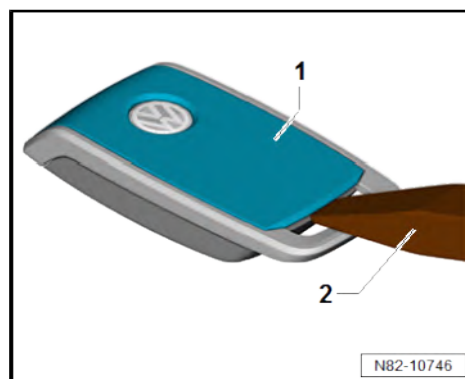
Special tools and workshop equipment required

- ◆ Removal wedge - 3409-



Note

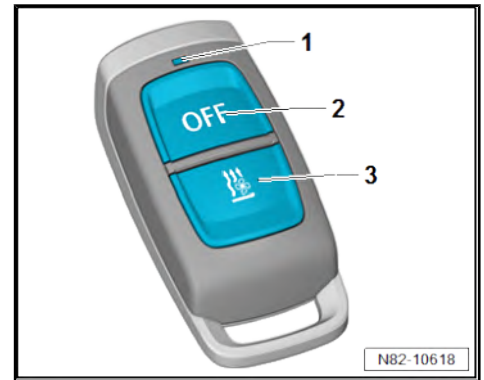
- ◆ *Note installation position of battery.*
 - ◆ *Only use batteries of the same type with a voltage of 3 volts.*
 - ◆ *Allocation of the remote control to the auxiliary/supplementary heater remains intact. Re-adapting is not necessary.*
- Release battery compartment cover -1- using removal wedge - 3409- -2-.
 - Remove battery.
 - Insert new battery and refit cover.





Indicator lamp in remote control

Once you have pressed a button, the indicator in the remote control -1- indicates whether the signal from the remote control has been received by the auxiliary heater and whether the battery in the remote control is low.



Display -1-	Meaning
LED lights up green for 2 seconds	The auxiliary heater has been switched on with button -3-.
LED lights up red for 2 seconds	The auxiliary heater has been switched on with button -2-.
Diode flashes green for 0.5 seconds 4 times	Signal to switch on was not received.
Diode flashes red for 0.5 seconds 4 times	Signal to switch on was not received.
Diode lights up green with following cycle 3 times: ♦ 0.65 on/0.2 off/0.15 on/0.5 off	Auxiliary heater is blocked, e.g. on account of fuel tank being nearly empty or a fault in the auxiliary heater.
LED lights up red for 2 seconds	Auxiliary heater is blocked, e.g. on account of fuel tank being nearly empty or a fault in the auxiliary heater.
Diode lights up orange for 2 seconds and then green for 2 seconds	Battery charge is low but switch on/switch off signal was received.
Diode lights up orange for 2 seconds and then red for 2 seconds	Battery charge is low but switch on/switch off signal was received.
Diode lights up orange for 2 seconds and then flashes green for 0.5 seconds 4 times	Battery charge is low, switch on/switch off signal was not received.
Diode lights up orange for 2 seconds and then flashes red for 0.5 seconds 4 times	Battery charge is low, switch on/switch off signal was not received.
Diode lights up with following cycle 3 times: ♦ 2 seconds orange then 0.65 on/0.2 off/0.15 on/0.5 off green	Auxiliary heater is blocked, e.g. on account of fuel tank being nearly empty or a fault in the auxiliary heater.
Diode lights up orange for 2 seconds and then red for 2 seconds	Auxiliary heater is blocked, e.g. on account of fuel tank being nearly empty or a fault in the auxiliary heater.
Flashes orange for 0.25 seconds 12 times	Battery is discharged, switch on/switch off signal was not received.
LED does not light up	Battery is discharged.

6.4 Removing and installing temperature sensor on roof for air inlet on auxiliary air heater - G967-



Note

Only on vehicle versions Kombi and panel van.



Special tools and workshop equipment required

- ◆ Removal wedge - 3409-

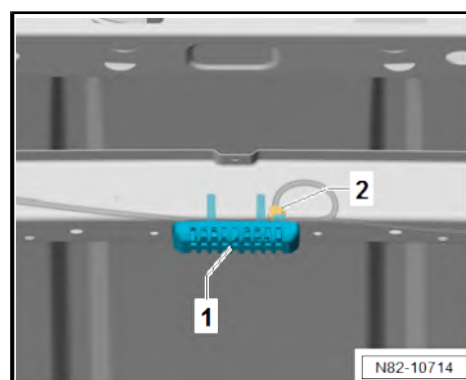


Removing

- Using removal wedge - 3409- , carefully lever temperature sensor on roof for air inlet on auxiliary air heater - G967- -1- out of roof cross member or moulded headliner.
- Disconnect electrical connector -2-.

Installing

Install in reverse order of removal.



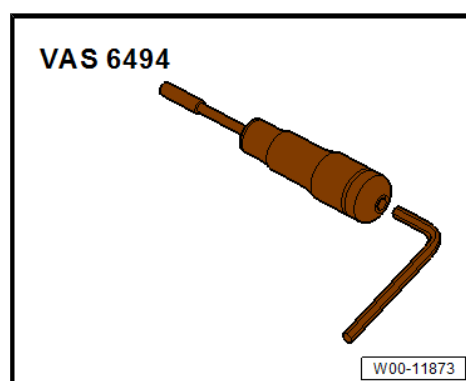
6.5 Removing and installing remote control receiver for auxiliary heater - R64-

See ➔ [page 68](#) for removal and installation.

6.6 Removing and installing air distribution flap control motor 2 - V667-

Special tools and workshop equipment required

- ◆ Torque screwdriver - VAS 6494-



- ◆ Vehicle diagnostic tester

Removing

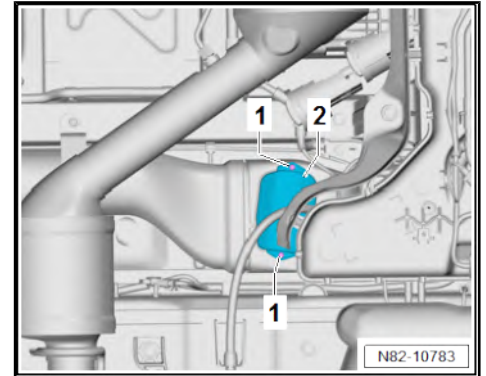
- Remove front left underbody cladding ➔ General body repairs, exterior; Rep. gr. 66 ; Underbody cladding; Removing and installing underbody cladding .



⚠ CAUTION

Danger of severe burns from hot exhaust system.
Burns to hands and other body parts possible.
– Allow exhaust system to cool.

- Remove bolts -1-.
- Remove cap -2-.

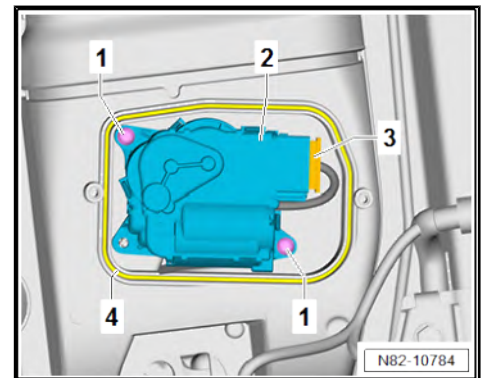


- Remove bolts -1-.
- Remove air distribution flap control motor 2 - V667- -2-. While doing this, disconnect electrical connector -3-.

Installing

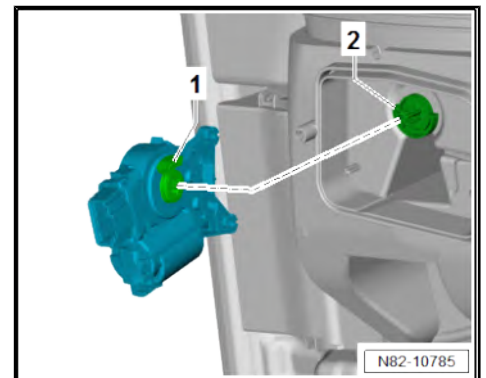
Install in reverse order of removal, observing the following:

- Inspect seal -4- for damage, and renew it if necessary ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.
- Check operation of flaps and hinge mechanism before fitting.
- Make sure levers and shafts are properly fitted in the mounts.



Note

- ◆ Mounting -1- on air distribution flap control motor 2 - V667- can be fitted on relay lever -2- in only one position.
- ◆ If mounting -1- on air distribution flap control motor 2 - V667- and relay lever -2- are not opposite each other, turn mounting -1- in air distribution flap control motor 2 - V667- .
- Initiate basic setting using ⇒ Vehicle diagnostic tester.
- Check function on auxiliary aid heater distribution box.



Specified torques

Component	Specified torque
Bolts on cap for air distribution flap control motor 2 - V667-	2.5 Nm
Bolts air distribution flap control motor 2 - V667-	1.5 Nm